

Bath dimension and initial entropy for closed repeated use of a quantum channel

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Abstract

We study finite closed devices that serve repeated uses of a fixed quantum channel while returning each output immediately and tolerating arbitrary adaptive inputs. One bath, one initial state and one unitary are fixed before the user; no reset, discard, fresh ancilla or uncounted controller is available. This differs from the fixed-memory exact single-use marginal repeatability of Rybár and Ziman: we allow horizon-dependent hardware and require vanishing complete error against adaptive users. Active dimension and actual initial entropy form distinct resources. Writing r for the limiting bath dimension per use, $\log R_T/T$, and s for the limiting initial entropy per use, $S(\omega_T)/T$, we characterize the rate region as $s \geq 0$, $r + s \geq h$ and $r - s \geq \kappa$, where h is maximum entropy exchange and κ is a smoothed independent-reference extension cost, an ordered limit that is not asserted to be computable, whose exact form at each fixed full-rank input is an affine transform of the zero-leakage quantum privacy funnel; the minimum dimension rate is $(h + \kappa)/2$. The proof combines entropy budgets, a uniform collision-gain formulation, active support repair, and a closed adaptive implementation of encoder-only fully quantum Slepian–Wolf recycling. The implementation counts the entire bath, including its seed, clock, workspace and permanently parked residues, and achieves actual initial entropy rates. The entropy-exchange functional, the privacy funnel, the minimax step and the Slepian–Wolf split are established ingredients; what is added here is the closed contract, a support repair that preserves the positive initial spectrum with a bath-independent gain penalty, and the counted adaptive implementation.

1 Introduction and model

A quantum channel is normally charged once. Fix a CPTP map Φ and a single use costs a dilation: an environment of dimension at most q^2 if it starts pure, or some other finite bath if it may start mixed. Serving T inputs in sequence is a different question, and the difference is not merely a factor of T : the closed cost is still linear in T , but its coefficient, and the trade between dimension and initial entropy, are what a single-use dilation does not determine. A device that has already produced outputs must have somewhere to keep what producing them left behind. If it may reset its bath between visits, discard a spent cell, or draw a fresh ancilla, the question collapses to the single-use one: use one dilation T times and let the rest arrive from outside. Those three permissions are exactly what a closed implementation lacks. Reset consumes a supply of pure states, discard needs a place to put what is discarded, and a fresh ancilla is a resource that somebody prepared. Charging all of them means fixing one finite Hilbert space and one finite initial state in advance and asking what fits inside.

Definition 1.1 is that contract. For each horizon T the device is one active bath B_T of dimension R_T , one user-independent initial state ω_T on it, and one unitary W_T applied at every visit; all three are chosen before the user and may depend on Φ and T . The output is handed back before the next input arrives. The tester is adaptive, may retain every output, and may hold arbitrary finite references, so the error of Definition 1.2 compares complete final states, not single-use marginals. Nothing inside is free: the seed, the clock, the workspace, the records and every register parked permanently after use are all part of R_T . The one object

outside is a purifier of ω_T , which the device may never touch and which is therefore never a resource.

Two numbers describe such a device, and they are not interchangeable. The first is $\log R_T$, the dimension it occupies. The second is $S(\omega_T)$, the actual entropy of the state it was initialized in. A rank count lies between them: $S(\omega) \leq \log \text{rank}(\omega) \leq \log R$, with equality throughout only for a flat positive spectrum, so neither dimension nor rank determines the other number. The four-dimensional example after Definition 1.4 separates all three. Initial entropy is the right second axis because the contract makes it a resource that can be spent: a device may be handed a mixed initializer, and what it is charged is the entropy of that initializer, not the provenance of the randomness, which lies outside this contract and is not identified here with work or energy. The region below then charges that convenience back. Past the corner, every further bit of initial entropy costs a bit of dimension, since $r \geq \kappa + s$ there.

Theorem 2.1 characterizes what is achievable. Writing r for the dimension rate and s for the actual initial entropy rate, the closed region is $s \geq 0$, $r + s \geq h(\Phi)$ and $r - s \geq \kappa(\Phi)$, and the least dimension rate is $(h + \kappa)/2$, attained with initial entropy rate $(h - \kappa)/2$ and a spectrum flat on its support. Both channel quantities are intrinsic. The first, h , is Schumacher's entropy exchange maximized over inputs. The second, κ , is an infimum of $S(\text{QA} | \text{Z})$ over finite extensions whose reference stays exactly in product with the auxiliary, smoothed in the output error, with the positive-error limit taken before the full-rank input supremum; Corollary 4.3 identifies the positive-error limit at each fixed full-rank input with $S(\chi_\rho) - P_q^{\psi\text{eq}}(0)$, an affine transform of the zero-leakage quantum privacy funnel. Kappa is the supremum of these limits. Two resources, two channel quantities, one closed region.

The closest antecedent contract is Rybár and Ziman's repeatable quantum memory channel: one fixed finite memory, initializer and unitary, required to produce the same single-use marginal at every repetition, for uncorrelated inputs. Rybár and Ziman also discuss finite n -repeatability and a finite-cell construction. Here we optimize dimension and actual initialization entropy as the horizon grows, while requiring vanishing complete error against arbitrary adaptive quantum users. The gap between the contracts is not cosmetic. One fair shared bath bit controlling $\mathbb{I}$ or Z is a legitimate marginal dephasing repeater in their sense, and Proposition 8.1 shows its complete service error is at least $1 - 2^{1-T}$, already $1/2$ at $T = 2$. No resolution of their original question is claimed here.

The contributions, in decreasing operational significance, are these. First, the characterization itself under the closed contract, together with its converse Theorem 2.2, which assumes no entropy-rate limit, constrains the same actual initialization through two separate legal experiments, and holds at every finite horizon with constants explicit in that horizon's own error. Second, the active support repair of Theorem 4.1: an approximate finite collision is corrected so that its Choi support lies inside the target's, with the positive initial spectrum unchanged and a gain penalty depending only on the fixed channel, not on the bath dimension, the initial spectrum or any auxiliary dimension; the subsequent flagged exactification changes the spectrum explicitly and says so, and with Theorem 3.1 it gives $\kappa_{\text{exact}} = \kappa$. Third, the implementation of Theorem 5.1 and Appendix A, which turns the encoder-only fully quantum Slepian–Wolf split into a device obeying Definition 1.1: spectral concentration uniform over adaptive controllers, one physical seed reused across every block and returned jointly independent of the whole future-accessed exterior, reversible flat-stock preparation on exact integer wires, and every residue allocated and counted in one closed bath. The entropy-exchange functional, the privacy funnel, the classical minimax step, the Slepian–Wolf split with its half-sum rates and the decoupling second moment are established ingredients, cited at their uses; the contract they are assembled under, and the repair and allocation needed to meet it, are the implementation tasks addressed here.

The characterization has real limits, stated where they arise. The quantity κ is an ordered limit of an infimum with no attainment, no bound on the auxiliary dimension, and no assertion

of computability or of continuity in Φ ; the thresholds in Section 6 are finite but not effective. Proposition 7.4 supplies a lower bound on κ that is an optimization over the input space alone, and Proposition 7.5 brackets a single-qubit channel with $0 < \kappa < h$, so the non-degenerate case $0 < \kappa < h$ occurs and the two degenerate edges are not the only cases. Nothing here bounds circuit size or latency, requires the bath to be returned, or extends beyond one fixed memoryless channel visited once per input.

Section 2 states the region and proves the converse. Section 3 identifies κ with an optimum over physical approximate collisions, Section 4 repairs such a collision into an exact one and derives the privacy-funnel form, and Section 5 with Appendix A builds an all-horizon device from one exact witness. Section 6 assembles the region from these, Section 7 gives worked channels and the limits above, and Section 8 compares the contract with neighbouring ones.

Definition 1.1 (closed causal device). Fix a CPTP map Φ from operators on $S = \mathbb{C}^q$ to operators on $A = \mathbb{C}^q$. For every integer horizon $T \geq 1$, a device consists of a finite active bath B_T of dimension R_T , a density operator ω_T on it, and a unitary $W_T : SB_T \rightarrow AB_T$, with S and A identified between visits. All three may depend on Φ and T and are chosen before the user. Initially ω_T is in product with the user's entire state. A mathematical purifier F is inaccessible forever; it is an analysis device only, the device never acts on it, so it is not charged, and the cost of a mixed ω_T enters through $S(\omega_T)$. Each visit applies this same W_T and hands A back before the next input arrives. All device seeds, clocks, stock, work wires, records and permanently parked residues are part of B_T . Offline preparation and unrestricted fixed hardware are allowed. No device reset, discard, fresh online ancilla, external randomness, purifier operation or uncounted controller is allowed. There is no bath-return requirement, marginal or independent.

Here *active bath* means the entire allocated device Hilbert space B_T , including registers parked permanently after use; the *active dimension* is R_T . The inaccessible purifier is excluded. A close antecedent is the unitary memory model of Rybár and Ziman [9, sections II–III]. Their infinite-repeatability definition fixes one finite device for exact single-use marginals with uncorrelated inputs; they also discuss finite n -repeatability and preallocated cells. Here we optimize horizon-dependent resources while approximating the complete process against adaptive inputs. Section 8 gives an explicit separation; no resolution of their original question is claimed.

Definition 1.2 (complete error and rates). A tester has arbitrary finite user memory and references, may retain all outputs, and applies arbitrary adaptive quantum operations between visits. Let $\rho_T^{\text{real}}(U)$ and $\rho_T^{\text{ideal}}(U)$ be its complete final states when connected to the device or to T independent Φ calls. With $D(\rho, \sigma) = \|\rho - \sigma\|_1/2$, set

$$\delta_T = \sup_U D(\rho_T^{\text{real}}(U), \rho_T^{\text{ideal}}(U)), \quad c_{\text{seq}}(\Phi) = \inf_{\delta_T \rightarrow 0} \limsup_{T \rightarrow \infty} \frac{\log R_T}{T}. \quad (1.1)$$

The supremum includes all finite reference sizes. *All-horizon* means a device for every positive integer T , with error tending to zero along all integers. An achievable finite rate pair has actual limits $\log R_T/T \rightarrow r$ and $S(\omega_T)/T \rightarrow s$; the rate region is the closure of these pairs. Thus $R_T = 2^{rT+o(T)}$: positive r means exponential growth, while $r = 0$ permits subexponential growth.

Logarithms and entropies are base two, $S(\rho) = -\text{Tr} \rho \log \rho$, with $0 \log 0 = 0$. An unconditional stopped experiment is covered by padding its continuation with dummy inputs. No bound conditional on a rare stopping outcome is promised. Batch access to all inputs would remove the immediate-output requirement and is a different task.

Definition 1.3 (intrinsic costs). For a minimal pure Stinespring dilation $V : \mathcal{S} \rightarrow \mathcal{AE}$ of Φ , write $\Phi^c(\rho) = \text{Tr}_{\mathcal{A}} V \rho V^*$ and $h(\Phi) = \max_{\rho} S(\Phi^c(\rho))$. Let $|\Omega_q\rangle = q^{-1/2} \sum_i |i, i\rangle$ and $J_{\Phi} = (\text{id} \otimes \Phi)(|\Omega_q\rangle\langle\Omega_q|)$, so that $\text{Tr} J_{\Phi} = 1$ and $\text{Tr}_{\mathcal{A}} J_{\Phi} = \mathbb{I}_{\mathcal{Q}}/q$. For $\rho > 0$ choose $|\psi_{\rho}\rangle = (\mathbb{I} \otimes \sqrt{q\rho})|\Omega_q\rangle$, whose $\mathcal{Q}$ marginal is ρ^T in this basis, and put $\chi_{\rho} = (\text{id} \otimes \Phi)(|\psi_{\rho}\rangle\langle\psi_{\rho}|)$. For $\varepsilon > 0$ define

$$k_{\rho,\varepsilon} = \inf S(\mathcal{QA} | \mathcal{Z})_{\sigma} \quad \text{subject to} \quad \sigma_{\mathcal{QZ}} = \rho^T \otimes \sigma_{\mathcal{Z}} \text{ exactly,} \quad D(\sigma_{\mathcal{QA}}, \chi_{\rho}) \leq \varepsilon, \quad (1.2)$$

and $\kappa(\Phi) = \sup_{\rho > 0} \lim_{\varepsilon \downarrow 0} k_{\rho,\varepsilon}$.

The quantity h maximizes the established entropy exchange of Schumacher [14, section V A, summary (iii), p. 2622]. Complementary dilations differ by environment isometries, so this number is independent of the minimal choice. In (1.2) the infimum runs over normalized states on $\mathcal{QAZ}$ and every finite $\mathcal{Z}$; there is no uniform dimension bound or attainment assertion. Here $S(X | Y) = S(XY) - S(Y)$. The value $k_{\rho,\varepsilon}$ is nonincreasing in ε and lies between $S(\rho) - \log q$ and $S(\chi_{\rho})$, so its positive-error limit exists and is finite. The positive-error limit is taken with ρ fixed, before the full-rank supremum. Equality with the exact-extension value at $\varepsilon = 0$ requires the stability argument after Theorem 4.1; it is not an assumption of this definition.

The exact fixed-input optimization is already related to a named entropy functional. Write $K_0(\rho)$ for the same infimum with $\sigma_{\mathcal{QA}} = \chi_{\rho}$ exactly. For $\psi_{\mathcal{QAE}} = (\text{id} \otimes V)\psi_{\rho}$, all finite $\mathcal{QA}$ extensions are obtained by a channel $\mathcal{E} \rightarrow \mathcal{Z}$, and the product constraint is $I(\mathcal{Q}; \mathcal{Z}) = 0$. Thus

$$K_0(\rho) = S(\chi_{\rho}) - P_{\mathcal{q}}^{\psi_{\mathcal{EQ}}}(0), \quad (1.3)$$

where $P_{\mathcal{q}}$ is the quantum privacy funnel of Datta, Hirche and Winter [15, section V, Eq. (22)], with their $X = \mathcal{E}$, $Y = \mathcal{Q}$, $R = \mathcal{A}$ and $W = \mathcal{Z}$. Its objective $I(YR; W)$ becomes $I(\mathcal{QA}; \mathcal{Z})$, and $S(\mathcal{QA} | \mathcal{Z}) = S(\chi_{\rho}) - I(\mathcal{QA}; \mathcal{Z})$. Their subscript $\mathcal{q}$ denotes quantum, not the dimension here. Supremum and infimum range over all finite auxiliary dimensions; neither attainment nor a dimension bound is asserted. Section 4 supplies the additional positive-error stability for this paper's fixed channel. The known functional and the Slepian–Wolf split credited in Section 5 are ingredients in the closed adaptive rate characterization proved below.

Definition 1.4 (finite collision). A finite unitary U on $\mathcal{SB}$ and a state $\tau_{\mathcal{B}}$ give $\Phi_C(\rho) = \text{Tr}_{\mathcal{B}} U(\rho \otimes \tau)U^*$ and the *active* bath channel $\Gamma_C(\rho) = \text{Tr}_{\mathcal{A}} U(\rho \otimes \tau)U^*$. Define $f_C(\rho) = S(\Gamma_C(\rho)) - S(\tau)$, $g_C = \max_{\rho} f_C(\rho)$ and $\kappa_{\text{exact}} = \inf_{\Phi_C = \Phi} g_C$. Here Γ_C excludes the inert purifier.

Channel distance is $D_{\diamond}(\Psi, \Phi) = \|\Psi - \Phi\|_{\diamond}/2$, equivalently the supremum half trace distance on reference-entangled inputs. The exact finite collision and its spectrum are fixed before invoking a block limit. Mixed-environment realizations and their unitary column constraints were studied by Terhal et al. [18, Eqs. (6)–(8)]. Repeated system–bath interactions also appear in Scarani et al. [26, Eqs. (1)–(2)]: their thermalization model supplies identically prepared bath qubits that each interact once with the same system. Our finite collision is one such unitary building block; its repeated-service implementation must allocate and count every cell within Definition 1.1's horizon-dependent closed bath.

For example, a four-dimensional bath initialized with eigenvalues $(3/4, 1/4, 0, 0)$ has log dimension 2, log rank 1, actual entropy $H_2(1/4) \approx 0.8113$, and entropy deficit $2 - H_2(1/4) \approx 1.1887$. Its minimal purifier dimension is 2, but those inert degrees of freedom are not available workspace. In general only $S(\omega) \leq \log \text{rank}(\omega) \leq \log R$ holds; equality of entropy and log rank requires a flat positive spectrum.

Conventions, register tables and the bibliography are collected in Appendix B. The proofs below use no theorem about finite-tracial closure and none about processes with more than one visit per input.

2 Main theorem and converse

Theorem 2.1 (rate region). *By Theorems 2.2, 3.1, 4.1 and 5.1, proved below, the rate region of Definition 1.2 is exactly*

$$s \geq 0, \quad r + s \geq h, \quad r - s \geq \kappa.$$

The minimum dimension rate is $(h+\kappa)/2$, attained with actual entropy $(h-\kappa)/2$ and a spectrum that is flat on its support. For $q = 1$ all intrinsic costs and the minimum rate vanish.

Theorem 2.2 (same-spectrum converse). *Every vanishing-error all-horizon family satisfies*

$$\liminf_{T \rightarrow \infty} \frac{\log R_T - S(\omega_T)}{T} \geq \kappa, \quad \liminf_{T \rightarrow \infty} \frac{\log R_T + S(\omega_T)}{T} \geq h.$$

No entropy-rate limit is assumed.

Proof. The entropy-decrease budget has the antecedent of Rybár and Ziman [9, section III, Eq. (3.7)]. The reference-conditioned telescope below strengthens its increment to the intrinsic extension cost. Fix a horizon simulator and its actual user-independent ω_T , with inert purifier F . Feed fresh purified copies of one full-rank ρ and retain $Y_t = Q_t A_t$. Before each visit $B_{t-1} F Y_{<t}$ is pure. The register Q_t is independent of $F Y_{<t}$, and tracing the active operation preserves this product exactly. Global purity and the service error δ_T give

$$S(B_t) - S(B_{t-1}) = S(Y_t | F Y_{<t}) \geq k_{\rho, \delta_T}, \quad \log R_T - S(\omega_T) \geq T k_{\rho, \delta_T}. \quad (2.1)$$

Explicitly, $Z_t = F Y_{<t}$ is finite and $Q_t Z_t$ is exactly product, since the visit acts only on its complement $S_t B_{t-1}$. The QA marginal is within δ_T of χ_ρ by the complete service guarantee and partial trace. Thus $Y_t Z_t$ is a feasible extension even though $\dim Z_t$ grows with T . The telescope starts at $S(B_0) = S(\omega_T)$ and ends at $S(B_T) \leq \log R_T$.

Two points about that extension each deserve a sentence. First, the service guarantee bounds the complete final state at horizon T , whereas the telescope reads the state just after visit t . The two agree here because this tester never touches $Y_{\leq t}$ again and every later visit acts on $S_{t'} B_{t'-1}$ alone, so the $Q_t A_t$ marginal is frozen from visit t onward and inherits the horizon- T bound. Second, Z_t contains the purifier F , which Definition 1.1 makes permanently inaccessible. That is not a use of a forbidden resource. The auxiliary Z of Definition 1.3 is a mathematical extension of a state, not a register that anybody operates, and $k_{\rho, \varepsilon}$ is an infimum over all such states, so the bound needs only that one finite state on $Q_t A_t Z_t$ with the two stated properties exists. No step here prepares, reads, measures or acts on F ; a device or tester that did so would violate Definition 1.1.

For each fixed ρ , take the positive-error limit and then the full-rank input supremum. Even if some δ_T vanish exactly, eventually $\delta_T \leq \varepsilon$ for each $\varepsilon > 0$, which supplies the required lower bound without identifying $k_{\rho, 0}$.

In a separate legal run, feed purified copies of an input attaining h . The ideal user entropy is Th , while purity and subadditivity give $S(Y^T) = S(B_T F) \leq \log R_T + S(\omega_T)$. Entropy continuity on the q^{2T} -dimensional user system gives

$$\log R_T + S(\omega_T) \geq Th - 2\delta_T T \log q - H_2(\delta_T). \quad (2.2)$$

Both testers constrain the same numerical initialization entropy because the device is fixed before either tester is chosen. Their final bath states can differ. For each $\zeta > 0$ the two eventual inequalities hold simultaneously, so adding them gives $\liminf \log R_T / T \geq (h+\kappa)/2$. This proves the converse part of Theorem 2.1, even if $S(\omega_T)/T$ oscillates. The entropy-continuity bound used here is Audenaert's [1, Theorem 1], in its eventual small-distance regime. The remaining implication is proved in Section 6. $\square$

3 Intrinsic gain and physical witnesses

Theorem 3.1 (uniform collision gain). *One has*

$$\kappa = \lim_{\eta \downarrow 0} \inf_{D_\diamond(\Phi_C, \Phi) \leq \eta} g_C.$$

Proof. Let $\mathcal{C}_\eta$ be the finite collisions within half-diamond distance η of Φ , and define $a_\eta(\rho) = \inf_{C \in \mathcal{C}_\eta} f_C(\rho)$ and $b_\eta = \inf_{C \in \mathcal{C}_\eta} \max_\rho f_C(\rho)$. These sets are nonempty: a pure minimal Stinespring isometry extends to a finite square collision.

Fix full-rank ρ and a feasible σ_{QAZ} . Purify it on QAZC. Its product QZ marginal has rank $q \text{rank}(\sigma_Z)$, so $\dim \mathbf{C} \geq \text{rank}(\sigma_Z)$. Purification uniqueness gives an isometry $\mathbf{S}\mathbf{B}_0 \rightarrow \mathbf{A}\mathbf{C}$ from $\psi_\rho \otimes \Omega_{\mathbf{B}_0\mathbf{Z}}$, where $\dim \mathbf{B}_0 = \text{rank}(\sigma_Z)$. Embed $\mathbf{B}_0$ into a bath of dimension $\dim \mathbf{C}$ and complete this isometry to a square active-only unitary. Initial positive eigenvalues are those of σ_Z ; the added dimensions have zero weight. The register Z remains inert. Purity gives $f_C(\rho) = S(\text{QA} \mid \text{Z})_\sigma$. In matching Schmidt coordinates, any normalized pure reference-input vector has the form $(X \otimes \mathbb{I})\psi_\rho$ with $\|X\|^2 \leq 1/\lambda_{\min}(\rho)$. Trace-norm contraction under this congruence gives $D_\diamond(\Phi_C, \Phi) \leq \varepsilon/\lambda_{\min}(\rho)$. Conversely a collision in $\mathcal{C}_\eta$ yields a feasible extension at error η . Therefore

$$a_{\varepsilon/\lambda_{\min}(\rho)}(\rho) \leq k_{\rho, \varepsilon} \leq a_\varepsilon(\rho). \quad (3.1)$$

Their positive-error limits coincide for each *fixed* ρ . No eigenvalue bound uniform in ρ is used. The Q marginal is ρ^T in a fixed canonical basis; its eigenvalues equal those of the physical input.

Finite convex combinations of gain *functions* are physical: $\mathbf{B} = \bigoplus_j \mathbf{B}_j$, $\tau = \bigoplus_j p_j \tau_j$ and $U = \bigoplus_j U_j$ give $f_C(\rho) = \sum_j p_j f_{C_j}(\rho)$ for all ρ , with channel error at most η if all branches lie in $\mathcal{C}_\eta$. Actual initial entropy is $H(p) + \sum_j p_j S(\tau_j)$, and active dimension is $\sum_j \dim \mathbf{B}_j$. The term $H(p)$ cancels in the gain because the same sector probabilities remain at the output. It continues to count in storage. The maximum of the mixed gain can be smaller than the average separate maxima.

For fixed η , every f_C is continuous and concave on the compact q -state space. Put $a = \sup_\rho a_\eta(\rho)$. Epsilon-optimal pointwise choices and a finite open subcover supply $f_1, \dots, f_N$ with $\min_j f_j(\rho) < a + \epsilon$ for every ρ . The downward set $\mathcal{D} = \{x : \text{some } \rho \text{ has } x_j \leq f_j(\rho) \text{ for all } j\}$ is closed by subsequence compactness of the input states and convex by concavity. The point $(a + \epsilon, \dots, a + \epsilon)$ is outside $\mathcal{D}$ by the finite cover. Strict finite-dimensional separation gives a normal with nonnegative coordinates: a negative coordinate would be unbounded above on a downward ray of $\mathcal{D}$. Normalize that nonzero normal to weights summing to one, giving $\max_\rho \sum_j p_j f_j(\rho) < a + \epsilon$. The physical flagged collision realizes this function, so weak minimax supplies $b_\eta = \sup_\rho a_\eta(\rho)$. There is no infinite mixture, bath compactness, attainment assumption or finite size bound.

Finally $S(\rho) - \log q \leq f_C(\rho) \leq S(\rho) + \log q$ by entropy inequalities on AB. These bounds are independent of the bath. The infimum a_η is concave, so $\rho_t = (1 - t)\rho + t\mathbb{I}/q$ satisfies $a_\eta(\rho_t) \geq a_\eta(\rho) - 3t \log q$. Thus at each η the full-rank supremum equals the all-input supremum, without assuming boundary continuity. Shrinking error sets increase their infima, and hence

$$\lim_{\eta \downarrow 0} b_\eta = \sup_{\eta > 0} \sup_{\rho > 0} a_\eta(\rho) = \sup_{\rho > 0} \sup_{\eta > 0} a_\eta(\rho) = \kappa. \quad (3.2)$$

Only two suprema have been interchanged. $\square$

This yields finite approximate collisions uniformly over inputs at every positive accuracy and gain slack. It yields no exact witness at zero error and no uniform bound on witness dimension or spectrum. Pointwise lifting uses purification uniqueness and the semilocalization mechanism of Eggeling, Schlingemann and Werner [2, section III]. After physical flags convexify

gain functions, the uniform step is an instance of classical minimax, as in Sion [16, Theorem 4.2 (Kneser–Fan), p. 175]. The compact side is the input state space, and evaluation is concave in the input and affine in the gain function. The finite separation proof above makes this application explicit. The error conversion and limit order then give the collision interpretation needed here.

4 Active repair and exactification

Theorem 4.1 (active support repair and exactification). *For a fixed Φ with Choi rank k , every sufficiently η -close finite collision (U, τ) on R bath dimensions admits a support-repaired collision on $(k+1)R$ dimensions with the same positive initial spectrum, Choi support contained in that of Φ , channel error at most $\eta + \varepsilon_\eta$, and gain increase at most $\beta_q(\varepsilon_\eta)$, where $\varepsilon_\eta = O_\Phi(\eta^{1/6})$. A further counted flagged correction implements Φ exactly, on at most $(k+1)R + k$ dimensions, with gain increase tending to zero uniformly in R and τ . Combining this repair with Theorem 3.1 gives $\kappa_{\text{exact}} = \kappa$. The small-error threshold and constants may depend on the fixed Φ (including k , C_Φ and its smallest positive Choi eigenvalue μ), but not on R , τ or the auxiliary dimension.*

Lemma 4.2 (amplified Gram completion). *The fixed Kraus span admits the following almost-isometry completion with an auxiliary-dimension-independent constant. Uniform amplification bounds for maps into a fixed matrix algebra are standard, including Smith’s lemma [28]. We give an explicit trace-slice bound and use it to complete the Gram defect within the prescribed Kraus span. Fix a minimal Kraus family A_a for Φ , $\mathcal{K} = \text{span } A_a$ and $\mathcal{D} = \text{span } A_a^* A_b$. The Gram map $L(E_{ab}) = A_a^* A_b$ maps $\mathbb{I}_k$ to $\mathbb{I}_q$. A Hermiticity-preserving right inverse $\mathcal{R}$ on $\mathcal{D}$ can be expressed as a finite sum of trace functionals times fixed Hermitian matrices. The slice inequality bounds every amplification $\mathcal{R} \otimes \text{id}$ by one constant C_Φ independent of the auxiliary dimension.*

Proof. Choose Hermitian bases H_l of $\mathcal{D}$, Hermitian preimages C_l , and trace-duals F_l . Then $\mathcal{R}(X) = \sum_l \text{Tr}(F_l X) C_l$ and one may take

$$C_\Phi = \max\left(1, \sum_l \|F_l\|_1 \|C_l\|_\infty\right). \quad (4.1)$$

Diagonalizing F_l expresses each amplified slice as a signed sum of diagonal compressions of X ; each compression has norm at most $\|X\|$. This proves the bound at every dimension. Positivity of $\mathcal{R}$ is neither asserted nor needed.

For $V = \sum_a A_a \otimes B_a$ with $\Delta = V^* V - \mathbb{I}$ and $\|\Delta\| \leq d$, set $t = C_\Phi d$ and $G = (t\mathbb{I} - (\mathcal{R} \otimes \text{id})(\Delta))/(1+t) \geq 0$. Its square-root block columns C_a obey $C_a^* C_b = G_{ab}$. Thus $W = \sum_a A_a \otimes C_a$ has $W^* W = (L \otimes \text{id})(G) = (t\mathbb{I} - \Delta)/(1+t)$. The stacked map $[V/\sqrt{1+t}; W]$ is therefore an exact isometry whose coefficients stay in $\mathcal{K}$. Polar normalization need not preserve $\mathcal{K}$ and is not used. Here the block columns C_a map the input bath $\mathbf{H}_0$ into $\mathbb{C}^k \mathbf{H}_0$, and $\|W\| \leq \sqrt{(t+d)/(1+t)}$. $\square$

Proof of Theorem 4.1. The coefficient projection below is an offline matrix construction; the device does not measure a Choi projector. Throughout this proof ϵ is a cutoff parameter, distinct from the smoothing error ε of Definition 1.3.

For an η -close mixed collision U, τ implementing Ψ , choose a bath eigenbasis $\tau = \sum_j \lambda_j |j\rangle\langle j|$ and write $U = \sum_{b,j} U_{bj} \otimes |b\rangle\langle j|$. If $\Pi_{\mathcal{K}}$ is Hilbert–Schmidt projection of system matrices onto $\mathcal{K}$, set $V = \sum_{b,j} \Pi_{\mathcal{K}}(U_{bj}) \otimes |b\rangle\langle j|$ and $L_0 = U - V$. The channel Kraus operators are $\sqrt{\lambda_j} U_{bj}$; the projection itself is unweighted and the spectrum enters their Choi sum through λ_j . Let $P_{\mathcal{K}}$ be the projector onto the vectorized Kraus span, the support of J_Φ . Normalized Choi leakage

is $p = \text{Tr}((\mathbb{I} - P_{\mathcal{K}})J_{\Psi}) = \text{Tr}((\mathbb{I} \otimes \tau)L_0^*L_0)/q \leq \eta$. Set $E = \text{Tr}_{\mathcal{S}} L_0^*L_0$. The positive-operator inequality $X \leq q\mathbb{I} \otimes \text{Tr}_{\mathcal{S}} X$ and the cutoff $P = \mathbf{1}_{[0, \epsilon^2/q]}(E)$ give

$$\|L_0(\mathbb{I} \otimes P)\| \leq \epsilon, \quad \text{Tr}(\tau(\mathbb{I} - P)) \leq q^2\eta/\epsilon^2. \quad (4.2)$$

For completeness the domination follows by applying Cauchy–Schwarz to the q vectors $X^{1/2}(|i\rangle\xi_i)$, then bounding each diagonal block X_{ii} by their sum. The cutoff-weight bound follows from $E \geq (\epsilon^2/q)(\mathbb{I} - P)$ and $\text{Tr}(\tau E) = qp$. Neither estimate assumes $[\tau, P] = 0$.

Repair V on SPB with the Gram construction, and use the target’s pure Stinespring map on $\mathcal{S}(\mathbb{I} - P)\mathcal{B}$, retaining the complete bad-bath input via $|\psi\rangle|b\rangle \mapsto \sum_a A_a|\psi\rangle|a\rangle|b\rangle$ in the sector $\mathbb{C}^k(\mathbb{I} - P)\mathcal{B}$, orthogonal to $\mathcal{B}$ and $\mathbb{C}^k P\mathcal{B}$ used by the good map. The resulting isometry has bath output $\hat{\mathcal{B}} = \mathcal{B} \oplus (\mathbb{C}^k \otimes \mathcal{B})$, of dimension $(k+1)R$. Initialize $\hat{\tau} = \tau \oplus 0$ and complete the initialized columns to an active-only unitary. The old positive spectrum is unchanged, the old $\mathcal{F}$ remains inert, and the system channel’s Choi support lies in the target face. Off-diagonal cutoff coherences are retained in the global output; no pinching, measurement or discard occurs.

With $d_\epsilon = 2\epsilon + \epsilon^2$, $t_\epsilon = C_\Phi d_\epsilon$ and $a_\epsilon = \epsilon + t_\epsilon/2 + \sqrt{(t_\epsilon + d_\epsilon)/(1 + t_\epsilon)}$, the joint purified output distance is at most $a_\epsilon + 2q\sqrt{\eta}/\epsilon$, uniformly over inputs. Here $\|\Delta\| \leq d_\epsilon$ on the good sector because $V(\mathbb{I} \otimes P) = U(\mathbb{I} \otimes P) - L_0(\mathbb{I} \otimes P)$ with U an isometry and $\|L_0(\mathbb{I} \otimes P)\| \leq \epsilon$, so $\|(\mathbb{I} \otimes P)(V^*V - \mathbb{I})(\mathbb{I} \otimes P)\| \leq 2\epsilon + \epsilon^2$. The good-input operator distance is at most a_ϵ : on that sector $V/\sqrt{1 + t_\epsilon} - U = (V - U)/\sqrt{1 + t_\epsilon} + (1/\sqrt{1 + t_\epsilon} - 1)U$ has norm at most $\epsilon + t_\epsilon/2$, since U is an isometry and $1 - 1/\sqrt{1 + t} \leq t/2$, while the W block contributes at most $\sqrt{(t_\epsilon + d_\epsilon)/(1 + t_\epsilon)}$. Taking $\epsilon = \eta^{1/3}$ gives $\varepsilon_\eta = O_\Phi(\eta^{1/6})$. The bad component of the same initial purification has vector norm at most $q\sqrt{\eta}/\epsilon$; two isometries differ on it by at most twice that norm. Pure-state half distance is bounded by vector distance. Thus cutoff coherences are controlled without removing them. At $\eta = 0$ retain the original collision. Both gains equal $S(\text{QA} \mid \mathcal{F})$ on a q -dimensional input purification. Thus the conditional-entropy bound of Winter [4, Lemma 2] gives $\sup_\rho |f_{\hat{C}} - f_C| \leq \beta_q(\varepsilon_\eta)$, where $\beta_q(\varepsilon) = 4\varepsilon \log q + (1 + \varepsilon)H_2(\varepsilon/(1 + \varepsilon))$, for sufficiently small $\varepsilon_\eta \leq 1$. The bound involves $\dim \text{QA} = q^2$; it has no bath or purifier dimension factor.

Let μ be the smallest positive eigenvalue of J_Φ and $\bar{\eta} = \eta + \varepsilon_\eta$. Inside its support, $\|J_{\hat{\Psi}} - J_\Phi\|_\infty \leq 2\bar{\eta}$ and $J_\Phi \geq \mu P_{\mathcal{K}}$. Therefore $f = 2\bar{\eta}/(\mu + 2\bar{\eta})$ ensures $J_\Phi - (1 - f)J_{\hat{\Psi}} \geq 0$. Its partial trace is $f\mathbb{I}/q$, so division by f defines a CPTP correction Θ of Kraus rank at most k . A counted direct-sum collision implements $\Phi = (1 - f)\hat{\Psi} + f\Theta$ exactly and obeys

$$g_{\text{exact}} \leq g_C + \beta_q(\varepsilon_\eta) + f \log k. \quad (4.3)$$

Indeed $g_{\text{exact}} \leq (1 - f)(g_C + \beta_q(\varepsilon_\eta)) + f \log k$, and $g_C \geq f_C(\pi_q) \geq 0$ by entropy conservation and subadditivity. This uses the function-level flag identity. At zero error the correction is omitted. Final active dimension is at most $(k+1)R + k$. If the original positive eigenvalues are λ_i , the exact collision’s eigenvalues are $((1 - f)\lambda_i, f)$, with entropy $H_2(f) + (1 - f)S(\tau)$. Thus spectrum preservation belongs to support repair; exactification changes the spectrum explicitly. The initial entropy itself need not change uniformly little as R grows. That last sentence describes the repair; it is not a gap in what the repair hands on. Theorem 5.1 accepts any fixed finite exact collision together with whatever finite spectrum that collision actually has. It nowhere asks the exact witness’s spectrum to be close to the approximate one’s, and it uses no bound on eigenvalue ratios, on bath dimension or on concentration constants. Section 6 selects a witness and fixes all of its dimensions, spectra and constants for each m before any horizon is chosen, so a spectrum that jumps at exactification costs nothing in the diagonal that follows.

Choose a sufficiently small fixed η for each gain slack, use Theorem 3.1 to choose one finite approximate witness, and exactify it. The uniform penalty gives $\kappa_{\text{exact}} \leq \kappa$; exact witnesses belong to every feasible set of Theorem 3.1, giving the reverse inequality. No horizon is selected in this argument. An exact pure Stinespring collision has gain h , so replacing any witness with

gain above h by this pure collision yields exact witnesses with $0 \leq g \leq h$ and $g \rightarrow \kappa$. These, with their actual arbitrary finite spectra, are the inputs Theorem 5.1 accepts. This closes the proof without using Theorem 5.1 or a horizon limit. $\square$

Corollary 4.3 (pointwise smoothing and privacy funnel). *Using the lifting construction in Theorem 3.1 and the pointwise repair in Theorem 4.1, for each fixed full-rank ρ one has*

$$\lim_{\varepsilon \downarrow 0} k_{\rho, \varepsilon} = K_0(\rho), \quad \kappa = \sup_{\rho > 0} [S(\chi_\rho) - P_q^{\psi_{\text{EQ}}}(0)].$$

Proof. To justify the exact mapping in Definition 1.3, purify any exact extension on QAZC. Uniqueness of purification gives an isometry from $\text{supp } \psi_E$ to ZC ; tracing C and extending the map arbitrarily outside that support gives a channel $E \rightarrow \text{Z}$. Conversely such a channel yields an extension, proving the claimed parametrization even when ψ_E is singular.

For stability, fix $\ell = \lambda_{\min}(\rho) > 0$. Any ε -feasible extension lifts to a finite collision C at error at most $\eta = \varepsilon/\ell$, with its objective equal to $f_C(\rho)$. The support repair above changes this pointwise gain by at most $\beta_q(\varepsilon_\eta)$. Let w_η be the final correction weight. Function-level flag mixing, $f_\Theta(\rho) \leq \log k$, and $f_C(\rho) \geq S(\rho) - \log q \geq -\log q$ give

$$f_{\text{exact}}(\rho) \leq f_C(\rho) + \beta_q(\varepsilon_\eta) + w_\eta(\log k + \log q).$$

The exact collision supplies an exactly feasible extension. Taking arbitrarily small optimization slack at each positive ε below the Φ -dependent threshold of Theorem 4.1, applied at $\eta = \varepsilon/\ell$, therefore gives

$$k_{\rho, \varepsilon} \leq K_0(\rho) \leq k_{\rho, \varepsilon} + \beta_q(\varepsilon_\eta/\ell) + w_{\varepsilon/\ell}(\log k + \log q). \quad (4.4)$$

The two penalties vanish at fixed Φ and ρ independently of the extension dimension. This proves the equality before taking the full-rank supremum. No infimum is interchanged with a limit, and no finite auxiliary-dimension bound, optimizer, or uniformity as ℓ tends to zero is asserted. The pointwise lower bound is needed here; $g_C \geq 0$ alone would not suffice. $\square$

This reformulation follows from this paper's repair, not from the prior privacy-funnel definition, and does not alter the order in Definition 1.3.

5 Causal balancing of a fixed exact collision

Theorem 5.1 (fixed exact witness). *Fix one finite exact collision $C = (U, \tau)$ for Φ , with full active cell dimension R . Write $\sigma = S(\tau)$, $v = \max_\rho S(\Gamma_C(\rho))$, $g = v - \sigma$ and suppose $g \leq h$. Then an all-horizon family in Definition 1.1 achieves vanishing complete adaptive error and actual rate limits*

$$r = b = \frac{h + g}{2}, \quad s = a = \frac{h - g}{2}.$$

Its complete initial positive spectrum is flat. More specifically it has a pure clock, a maximally mixed seed of dimension J_T , K_T mixed data qubits, and pure remaining data, with $\log J_T/T \rightarrow 0$ and $K_T/T \rightarrow a$ separately. No uniform bound on the fixed witness's dimension, eigenvalue ratios, concentration constants or circuit complexity is imposed.

Proof. Feeding π_q and using total entropy conservation gives $v \geq \sigma$. For a purified input, BF is a complementary output and F has entropy σ ; hence $h \leq v + \sigma$. Therefore $0 \leq a \leq \sigma$, $b = a + g$ and $v \leq \log R$. These are the only resource inequalities used to allocate the construction.

Appendix A supplies the complete proof. Lemma A.1 gives uniform adaptive spectral concentration. Lemma A.2 transfers the virtual minimal-environment support to a purification of the complete exterior and retains the normalization in the simultaneous projector bound.

Lemma A.3 proves the second moment with the physical seed still present. Lemma A.4 constructs the nonflat cell marginal reversibly from flat stock on exact integer wires. Proposition A.5 proves joint recycling and every prefix's error with disjoint parked residues. Proposition A.6 counts the clock, actual spectrum and every-horizon limit. All are proved in this paper. $\square$

The encoder is the sender's unitary from fully quantum Slepian–Wolf: sender = C, retained share = M, message = G, receiver = N and reference = H; the receiver is written N here to keep F for the purifiers of Definition 1.1 and Appendix A. Its decoder acts on GN, so omitting that decoder leaves MH unchanged. The physical device parks and counts G. The split and its half-sum rates are established prior ingredients, specifically Abeyesinghe, Devetak, Hayden and Winter [7, Theorems IV.1–IV.2, Lemma IV.5 and Eq. (30), pp. 6–9, with the asymptotic rate choice following Eq. (35) in section VII]. The present appendix supplies the adaptive spectral control, reversible flat-stock preparation and counted reuse of seed and workspace needed to turn this split into the stated closed device. These implementation steps are part of the operational synthesis.

6 Assembling the region

Proof of Theorem 2.1. The argument uses Theorems 2.2, 3.1, 4.1 and 5.1.

First apply the converse bounds of Theorem 2.2 to an arbitrary all-horizon family, writing $x_T = \log R_T/T$ and $y_T = S(\omega_T)/T$. For each $\zeta > 0$, eventually *both* $x_T - y_T \geq \kappa - \zeta$ and $x_T + y_T \geq h - \zeta$. Their common actual initialization therefore gives $2x_T \geq h + \kappa - 2\zeta$, regardless of whether y_T converges. Thus $\liminf x_T \geq (h + \kappa)/2$ and c_{seq} has this lower bound. For rate-limit pairs, $s \geq 0$, $r + s \geq h$ and $r - s \geq \kappa$ follow and survive closure. The definition gives $0 \leq \kappa \leq h$: $S(\text{QA} \mid \text{Z}) \geq S(\rho) - \log q$ is nonnegative at $\rho = \mathbb{I}/q$, while the exact trivial-Z extension has entropy $S(\chi_\rho) \leq h$.

Theorems 3.1 and 4.1 give $\kappa_{\text{exact}} = \kappa$. For each $m \geq 1$ select a finite *exact* collision with gain $< \kappa + 1/m$. Replace it by a pure exact Stinespring collision of gain h if its gain exceeds h . Then $\kappa \leq g_m \leq h$ and $0 \leq g_m - \kappa < 1/m$. This also handles $\kappa = h$ without assuming attainment. Set $a_m = (h - g_m)/2$ and $b_m = (h + g_m)/2$. All dimensions, spectra and concentration constants are fixed separately for each m .

Theorem 5.1 supplies, for each fixed witness, all-horizon devices with error $\delta_{m,T} \rightarrow 0$ and rates (b_m, a_m) . Their exact initialization is

$$\begin{aligned} \omega_{m,T} &= |0\rangle\langle 0|_{\text{clock}} \otimes \pi_{J_{m,T}} \otimes \pi_{2^{K_{m,T}}} \otimes |0\rangle\langle 0|_{\text{rest}}, \\ R_{m,T} &= (T+1) J_{m,T} 2^{A_{m,T}}, \quad S(\omega_{m,T}) = \log J_{m,T} + K_{m,T}. \end{aligned} \tag{6.1}$$

Here $0 \leq K_{m,T} \leq A_{m,T}$ are integers. Theorem 5.1 proves separately that $\log J_{m,T}/T \rightarrow 0$ and $K_{m,T}/T \rightarrow a_m$. Choose a finite H_m such that for *every* $T \geq H_m$, the error and each of

$$\left| \frac{\log R_{m,T}}{T} - b_m \right|, \quad \left| \frac{S(\omega_{m,T})}{T} - a_m \right|, \quad \frac{\log J_{m,T}}{T}, \quad \left| \frac{K_{m,T}}{T} - a_m \right|$$

are at most $1/m$. Take $T_1 = \max(1, H_1)$ and $T_{m+1} = \max(H_{m+1}, T_m + 1)$. For $T \geq T_1$ select the complete m -th device with $m = m(T) = \max\{j : T_j \leq T\}$; use exact pure preallocated cells for earlier T . Since $T_m \geq m$, this maximum is finite; for every M , $T \geq T_M$ implies $m(T) \geq M$. Thus $m(T) \rightarrow \infty$ over all horizons. Every selected T lies beyond H_m , so the bounds hold throughout $T_m \leq T < T_{m+1}$. With

$$r_0 = \frac{h + \kappa}{2}, \quad s_0 = \frac{h - \kappa}{2}, \tag{6.2}$$

the actual rate errors are at most $3/(2m(T))$, the adaptive error is at most $1/m(T)$, $K_T/T \rightarrow s_0$, and $\log J_T/T \rightarrow 0$. The separate seed threshold is needed for the region proof; it must not be inferred merely from total-rate limits. Hardware and its actual initializer are selected before the user for each T ; no witness changes online. Each elementary collision implements *exact* Φ , so there is no accumulated $T\eta$ term. This proves $c_{\text{seq}} = r_0$ with actual entropy rate s_0 . No effective thresholds or witness-size bound are claimed for this achievability half. The converse half is the half that constrains an actual device: for every finite horizon, Theorem 2.2's two inequalities hold with explicit constants in terms of that horizon's own error δ_T , although the extension-cost term k_{ρ, δ_T} itself is an infimum that is not asserted to be computable. Only the thresholds H_m above are non-effective.

The selected base family has $P_T = J_T 2^{K_T}$ equal positive eigenvalues $1/P_T$ and $R_T - P_T$ zeros. To reach $0 \leq s \leq s_0$ put $d = s_0 - s$ and $l_T = \min(K_T, \lfloor dT \rfloor)$. Then $l_T/T \rightarrow d$, including the clipped endpoint $s = 0$. Replace l_T of the initially mixed data qubits by halves of Bell pairs with l_T *new*, counted active partner qubits P . Keep the other initial factors unchanged and extend the repeated unitary to $W_T \otimes \mathbb{I}_P$. The new old-bath marginal is *exactly* ω_T . Partial trace over P commutes with every device and adaptive-user operation, so the entire original marginal process and its complete adaptive error are unchanged. This uses a new offline initializer, not an operation on the original inaccessible purifier.

Exactly, the new dimension is $R_T 2^{l_T}$, its positive rank is $P_T^- = J_T 2^{K_T - l_T}$, and its positive spectrum is $1/P_T^-$ repeated P_T^- times, with zeros elsewhere. Its entropy is $S(\omega_T) - l_T$. Therefore the limiting pair is $(r_0 + d, s_0 - d) = (h - s, s)$. At $s = 0$ the remaining entropy is $\log J_T + \max(K_T - \lfloor s_0 T \rfloor, 0) = o(T)$; a zero entropy *rate* does not require exactly pure finite-horizon initialization. Partners remain counted and untouched, and a purifier of the new initializer remains inaccessible.

For $s \geq s_0$ instead append $n_T = \lfloor (s - s_0)T \rfloor$ unused maximally mixed qubits in product, extending W_T by identity. Dimension and positive rank both multiply by 2^{n_T} and entropy increases by n_T . The positive spectrum remains flat and the user process is unchanged. This gives $(\kappa + s, s)$. For any $r \geq r_{\min}(s) = \max(h - s, \kappa + s)$, append $\lfloor (r - r_{\min}(s))T \rfloor$ unused pure qubits. This increases dimension, leaves positive rank and entropy unchanged, and adds only zero eigenvalues. All rounding errors vanish after division by T .

Every finite rate pair in the stated region is thus realized with actual rate limits; the set is already closed. At each finite horizon, $\text{rank} \leq \text{dimension}$ and $S \leq \log R$ hold: internal purification decreases rank and increases dimension, mixed padding multiplies both equally, and pure padding only increases dimension. Asymptotically $r \geq s$ follows from $\kappa \geq 0$. At $q = 1$ the trivial bath-free device and padding realize $r \geq s \geq 0$. If $\kappa = h$ then $s_0 = 0$; if $\kappa = 0$ then $r_0 = s_0 = h/2$. Both endpoints are covered by the same integer constructions.

The dependency order is acyclic: Theorem 2.2 is an independent converse; the fixed-error minimax of Theorem 3.1 precedes the exactification of Theorem 4.1; Theorem 5.1 acts on one fixed exact witness without Theorems 3.1 and 4.1; the all-horizon diagonal then combines these inputs. Internal purification and padding are last. The encoder-only Slepian–Wolf attribution of Section 5 remains in force. $\square$

7 Examples and limitations

Proposition 7.1 (unitary channels). *For $\Phi(\rho) = V\rho V^*$, $h = \kappa = 0$ and the minimum memory rate is zero.*

Proof. A minimal environment is one-dimensional, so $h = 0$. Every feasible extension has $S(\text{QA} | \text{Z}) \geq S(\rho) - \log q$, which is nonnegative at $\rho = \pi_q$. Taking a trivial Z gives $\kappa \leq h$, hence $\kappa = 0$. The bath-free repeated V implements all visits exactly and has both rates zero. $\square$

Proposition 7.2 (qubit dephasing). *For $\Delta(\rho) = (\rho + Z\rho Z)/2$, $h = 1$, $\kappa = 0$, and Theorem 2.1 gives the corner $r = s = 1/2$.*

Proof. A minimal dilation copies the computational-basis label into a qubit environment. Its output entropy is the binary entropy of the input diagonal, maximized at one. A fair classical bath bit controlling $\mathbb{I}$ or Z implements Δ . Its active marginal stays π_2 for every input, so $g = 0$. Theorem 3.1 and $\kappa \geq 0$ imply $\kappa = 0$. $\square$

There is also a direct exact streaming realization of the corner. Put $m = \lceil T/2 \rceil$, initialize m bath qubits in π_{2^m} , and choose T distinct binary-independent Pauli generators from $X_1, Z_1, \dots, X_m, Z_m$. At visit t use the system computational bit to control the t -th generator P_t on the bath. For a history $x \in \{0, 1\}^T$, its ordered bath word is $P(x) = P_T^{x_T} \cdots P_1^{x_1}$. Distinct histories give distinct Pauli words up to phase and $\text{Tr}(P(x)P(y)^*)/2^m = \delta_{x,y}$. Expanding any purified adaptive tester in computational histories, its retained user vectors are weighted by precisely this Gram matrix after tracing the bath. All cross-history terms vanish, exactly as for fresh dephasing records; this proves complete adaptive service, including reference-entangled inputs. A pure $(T+1)$ -state clock compiles these gates into one repeated unitary as in (A.5). Thus $\log R = m + \log(T+1)$ and $S(\omega) = m$, with no discard. This is the counted streaming specialization of the orthogonal-unitary construction in Boes et al. [8, Lemma 1], not a claim to originate its square-root dephasing memory saving. The subsequent 2020 erratum corrects section V / Theorem 3's expander result, which is not used here; it does not modify the dephasing lemma cited above.

Proposition 7.3 (pure replacement). *For $\Phi(\rho) = |0\rangle\langle 0| \text{Tr } \rho$ on $\mathbb{C}^q$, $h = \kappa = \log q$ and $c_{\text{seq}} = \log q$.*

Proof. A minimal dilation maps $|\psi\rangle$ to $|0\rangle_A |\psi\rangle_E$, so $h = \log q$. For any feasible extension, $S(\text{QA} | \text{Z}) \geq S(\rho) - S(\sigma_A)$. Its A marginal is within ε of $|0\rangle\langle 0|$, hence $S(\sigma_A) \rightarrow 0$ at fixed q by entropy continuity. At $\rho = \pi_q$ this gives $\kappa \geq \log q$, while $\kappa \leq h$ gives equality. Specializing the preallocated cell construction of Rybár and Ziman [9, section III], preallocate T pure q -dimensional cells and swap each arriving input into its designated cell, returning $|0\rangle$; retain all cells and count the pure clock. This exact device has $\log R = T \log q + \log(T+1)$ and $S(\omega) = 0$. Its bath grows with T , as it must: Theorem 2 of Rybár and Ziman excludes a fixed finite memory for this nonunital channel. No mixed-replacement or other companion classification is needed for this example. $\square$

Proposition 7.4 (a lower bound on κ over the input space alone). *For every channel Φ ,*

$$\kappa(\Phi) \geq \max_{\rho} [S(\rho) - S(\Phi(\rho))].$$

Proof. Fix $\rho > 0$ and any ε -feasible extension σ in Definition 1.3, with $q \geq 2$. The chain rule and the conditional form of Araki–Lieb give

$$S(\text{QA} | \text{Z}) = S(\text{Q} | \text{Z}) + S(\text{A} | \text{QZ}) \geq S(\rho) - S(\sigma_A),$$

using $S(\text{Q} | \text{Z}) = S(\rho)$ from the exact product $\sigma_{\text{QZ}} = \rho^T \otimes \sigma_Z$ and $S(\text{A} | \text{QZ}) \geq -S(\text{A})$. Since $D(\sigma_{\text{QA}}, \chi_\rho) \leq \varepsilon$ and $\text{Tr}_Q \chi_\rho = \Phi(\rho)$, the marginal σ_A is within half distance ε of $\Phi(\rho)$, so Audenaert [1, Theorem 1] gives $S(\sigma_A) \leq S(\Phi(\rho)) + \varepsilon \log(q-1) + H_2(\varepsilon)$. Hence $k_{\rho, \varepsilon} \geq S(\rho) - S(\Phi(\rho)) - \varepsilon \log(q-1) - H_2(\varepsilon)$. Letting $\varepsilon \downarrow 0$ at fixed ρ and then taking the supremum over full-rank ρ gives $\kappa \geq \sup_{\rho > 0} [S(\rho) - S(\Phi(\rho))]$, which equals the maximum over all states by continuity of the bracket and density of the full-rank states; the maximum exists by compactness. This uses neither Theorem 3.1 nor Theorem 4.1.

The same bound holds for every finite *exact* collision $C = (U, \tau)$, by the entropy-conservation and subadditivity budget of Rybár and Ziman [9, section III, Eq. (3.7)]: the state

$U(\rho \otimes \tau)U^*$ on $\mathbf{AB}$ is unitarily equivalent to $\rho \otimes \tau$, so $S(\mathbf{AB}) = S(\rho) + S(\tau)$, while subadditivity gives $S(\mathbf{AB}) \leq S(\Phi(\rho)) + S(\Gamma_C(\rho))$. Subtracting $S(\tau)$,

$$f_C(\rho) \geq S(\rho) - S(\Phi(\rho)) \quad \text{for every } \rho,$$

hence $g_C \geq \max_\rho [S(\rho) - S(\Phi(\rho))]$. This second route bounds κ_{exact} directly and reaches κ only through Theorems 3.1 and 4.1. $\square$

Unlike Definition 1.3, the right-hand side is an optimization over the q -dimensional state space alone: no auxiliary system, no smoothing and no order of limits enter it. It is exactly the universal parallel work rate at trivial Hamiltonians of Faist, Berta and Brandão [20, Theorem 5.1 and Eq. (5.4)], so the entropy-deficit rate of a closed device is at least that thermodynamic work rate; Section 8 compares the two contracts. The bound reproduces the three examples above exactly: it is zero for a unitary channel, zero for dephasing because a unital channel never decreases entropy, and $\log q$ for the pure replacer.

Proposition 7.5 (an interior corner). *For $0 \leq \nu, p \leq 1$ let $\Phi_{\nu,p}$ on $q = 2$ have the generalized amplitude damping operators*

$$\begin{aligned} K_0 &= \sqrt{p} \text{diag}(1, \sqrt{1-\nu}), & K_1 &= \sqrt{p\nu} |0\rangle\langle 1|, \\ K_2 &= \sqrt{1-p} \text{diag}(\sqrt{1-\nu}, 1), & K_3 &= \sqrt{(1-p)\nu} |1\rangle\langle 0|, \end{aligned}$$

which satisfy $\sum_a K_a^* K_a = \mathbb{I}$; the damping parameter is written ν to keep γ for the concentration slack of Appendix A. At $\nu = 1/2$ and $p = 3/4$,

$$0 < H_2(7/12) - H_2(1/3) \leq \kappa \leq 1 - H_2(3/4) < H_2(3/8) \leq h,$$

that is $0.061572 \leq \kappa \leq 0.188722$ and $h \geq 0.954434$, each decimal rounded towards the side that keeps the inequality valid. Hence $0 < \kappa < h$.

Proof. Trace preservation is the diagonal identity $p + (1-p)(1-\nu) + (1-p)\nu = 1$ and $p(1-\nu) + p\nu + (1-p) = 1$. For the lower bound, the channel acts on diagonal inputs by $\Phi(\text{diag}(a, 1-a)) = \text{diag}(3/8 + a/2, 5/8 - a/2)$, so $\rho = \text{diag}(7/12, 5/12)$ has output exactly $\text{diag}(2/3, 1/3)$, and Proposition 7.4 at this ρ gives $\kappa \geq H_2(7/12) - H_2(1/3) > 0$ through the entropy-decrease budget credited there. The maximally mixed input, with $\Phi(\mathbb{I}/2) = \text{diag}(5/8, 3/8)$, gives only $1 - H_2(5/8) = 0.04556\dots$ and is not the maximiser of the bracket in Proposition 7.4; N. Mghirbi (private communication, 2026) observed that $\text{diag}(3/5, 2/5)$ already improves it to 0.061214. A numerical search over qubit inputs, rerun by the script named in Appendix B.4, finds the largest value 0.061598 at $\text{diag}(0.58671, 0.41329)$, so the rational witness above is within 0.00003 of the largest value found numerically; the search is a diagonal grid with random general inputs and certifies no global maximum.

For the upper bound, exhibit an exact collision. Take $\mathbf{B} = \mathbb{C}^2$, $\tau = \text{diag}(p, 1-p)$, and the unitary U fixing $|00\rangle$ and $|11\rangle$ and acting on the ordered pair $|01\rangle, |10\rangle$ by

$$\begin{pmatrix} \sqrt{1-\nu} & -i\sqrt{\nu} \\ -i\sqrt{\nu} & \sqrt{1-\nu} \end{pmatrix}.$$

That block is unitary, so U is. Writing $U = \sum_{b,j} U_{bj} \otimes |b\rangle\langle j|$ as in Section 4, its weighted bath matrix elements $\sqrt{\lambda_j} U_{bj}$ come from $U_{00} = \text{diag}(1, \sqrt{1-\nu})$, $U_{11} = \text{diag}(\sqrt{1-\nu}, 1)$, $U_{10} = -i\sqrt{\nu}|0\rangle\langle 1|$ and $U_{01} = -i\sqrt{\nu}|1\rangle\langle 0|$, weighted by $\sqrt{p}$ and $\sqrt{1-p}$; these are the four operators above up to individual phases, so $\Phi_C = \Phi$ exactly. This is a lawful finite collision in the sense of Definition 1.4: τ is user independent, U acts on system and active bath only, and $\log \dim \mathbf{B} = 1$ with $S(\tau) = H_2(p)$. Because $\dim \mathbf{B} = 2$, its active output entropy is at most one for *every* input, whatever the off-diagonal entries, so $g_C \leq 1 - H_2(3/4)$. This bound is attained: the

input $\text{diag}(1/4, 3/4)$ sends the active bath output of this collision to $\mathbb{I}/2$, so $g_C = 1 - H_2(3/4)$ exactly and the witness cannot be improved. Theorems 3.1 and 4.1 give $\kappa = \kappa_{\text{exact}} \leq g_C$.

For h , the minimal dilation of a pure input has system and environment with the same nonzero spectrum, so $h \geq S(\Phi(|1\rangle\langle 1|)) = H_2(3/8)$, since $\Phi(|1\rangle\langle 1|) = \text{diag}(3/8, 5/8)$. A mixed input does better: at $\text{diag}(1/3, 2/3)$ the complementary output has entropy 1.148402 (also observed by N. Mghirbi; the largest value found numerically is 1.148986 at $\text{diag}(0.35586, 0.64414)$), so in fact $h > 1$, although the analytic bound suffices here. Strictness follows from H_2 being strictly increasing on $(0, 1/2)$: $H_2(3/8) > H_2(1/4) = H_2(3/4) > 1/2 > 1 - H_2(3/4)$. $\square$

This locates the corner without computing κ . Two consequences are already visible from the bracket. Since $\kappa > 0$, no device for this channel has a vanishing entropy-deficit rate, so $r = s$ is impossible and the minimum dimension rate $(h + \kappa)/2$ strictly exceeds $h/2$. Since $\kappa < h$, the optimal initial entropy rate $s_0 = (h - \kappa)/2$ is strictly positive, so the cheapest device is strictly mixed rather than pure. The bracket also gives $g_C \leq h$ for this witness, so Theorem 5.1 applies to it directly and produces an explicit all-horizon family. The finite arithmetic behind the displayed decimals is rerun by the script named in Appendix B.4; the strict inequalities above are analytic and use no optimizer.

The quantity c_{seq} measures complete operating storage. It is not identified with a universal unavoidable terminal residual rate under unrestricted growth of a working bank. A construction's parked-residue rate does not establish such an equality. The theorem asserts no efficient witness search, circuit bound, bath return, independent return, spatial additivity, or service law for arbitrary multi-visit processes. It concerns one fixed memoryless channel.

8 Related work and distinct zero-cost statements

Rybár and Ziman's *Repeatable quantum memory channels* [9, sections II–IV] is a close physical comparison. After Eq. (2.3) they assume product inputs. Their section III repeatability definition fixes one memory, initializer and unitary for all repetitions and requires the same single-use marginal channel at every step. Theorem 1 supplies finite random-unitary repeaters; Theorem 2 rules out nonunital finite-memory repeaters. Their conclusion leaves general unital, non-random-unitary repeatability open and acknowledges correlations between outputs and effects of measurements. These are statements of that 2008 paper, not a claim about the current status of its question.

Proposition 8.1 (shared-seed separation). *One fair bath bit controlling $\mathbb{I}$ or Z on every qubit is such a marginal dephasing repeater, but its complete service error against independent dephasing is at least $1 - 2^{1-T}$.*

Proof. For any product input list, the bath bit's probabilities remain $(1/2, 1/2)$, so each individual output has dephasing's marginal. For T inputs $|+\rangle$, the joint output is $(|+\rangle\langle +|^{\otimes T} + |-\rangle\langle -|^{\otimes T})/2$. Independent dephasing produces π_{2^T} . Both states are diagonal in the X basis. Summing the two occupied-string discrepancies and the other $2^T - 2$ discrepancies gives half distance $1 - 2^{1-T}$. The event that all X outcomes agree attains it, already $1/2$ for $T = 2$. No feedback is needed to distinguish the contracts. In particular, the finite shared seed is not a small-error device for Definition 1.2. $\square$

Our device may depend on T and has vanishing complete adaptive error. Those quantifiers and its joint service requirement differ from fixed finite exact marginal repeatability. Finite n -repeatability and the n -cell permutation construction are also in Rybár–Ziman; horizon dependence alone is not a new distinction. Theorem 2.1 is *not* asserted to resolve Rybár–Ziman's question or an unspecified asymptotic adaptive version.

The resource distinctions can be stated precisely from the core theorem.

Statement	Meaning and consequence
Zero initial entropy	$S(\omega_T) = 0$ at a specified finite horizon; stronger than its entropy being sublinear.
Zero initial entropy rate	$s = 0$; the region requires $r \geq h$. Exact purity at every horizon is not implied.
Zero entropy deficit	$\log R_T - S(\omega_T) = 0$ at a specified horizon, equivalently $\omega_T = \pi_{R_T}$.
Zero entropy-deficit rate	$r - s = 0$; under Theorem 2.1 this is possible iff $\kappa = 0$, with $r = s \geq h/2$.
Zero memory rate	$r = 0$; then $s = 0$ and $h = 0$. In the square-channel setting this is possible exactly for unitary channels.
Exact finite tracial factorization	One exact finite weighted-tracial dilation exists, with fixed finite algebra and weights.
Finite-tracial approximation closure	Arbitrarily accurate finite tracial dilations exist; no bounded dimension or exact attained dilation follows.

For the unitary characterization in this table, if $h = 0$ every complementary output is pure. Convexity forces these outputs to be the same pure state, since a mixture of distinct pure states has positive entropy. The dilation therefore factors as an isometry $S \rightarrow A$ tensor that pure state. Equal input and output dimensions make the isometry unitary; Proposition 7.1 gives the converse.

A finite weighted-tracial dilation means $B = \bigoplus_j \mathbb{C}^{d_j}$, $\tau = \bigoplus_j p_j \pi_{d_j}$, and a unitary preserving these sectors. It implements $\sum_j p_j \text{Tr}_B U_j (\rho \otimes \pi_{d_j}) U_j^*$. Each sector's output bath entropy is at most $\log d_j$ and π_q attains equality in every sector, so $g = 0$ by the block entropy identity. Theorem 3.1 therefore implies that membership in this approximation closure gives $\kappa = 0$. The reverse implication requires an additional gain-to-tracial approximation argument and is not a consequence proved in this paper.

Allowing weighted direct sums is a broader exact convention than a single maximally mixed matrix bath. Their approximation closures coincide: for large D choose nonnegative integers m_j with $\sum_j m_j d_j = D$ and $m_j d_j / D$ approaching p_j (for example take D along multiples of the d_j , and approximate the weights rationally). Replicate U_j on m_j copies. The half-diamond channel discrepancy is at most the total variation in weights. This observation supplies approximation, not exact attainment for irrational weights.

The distinction is substantive prior mathematics. Musat and Rørdam [10, Theorem 4.1] give Schur channels in the finite-factorization closure with no exact finite-dimensional factorization, in dimensions $2n+1$ for $n \geq 5$. We cite that theorem for this distinction, without re-proving its operator-algebraic dependency chain. Likewise Lie, Son, Boes, Ng and Wilming [11, published Proposition 6 and Eq. (12)] relate informational equilibrium to weighted-tracial dilations and factorizable channels. These structural and equilibrium ingredients are prior work. No companion finite-tracial-closure equivalence or quantitative modulus is imported into the results proved here.

Pointwise product-reference lifting uses the Stinespring uniqueness mechanism in Eggeling, Schlingemann and Werner [2, section III]. The uniform optimization and smoothing order in Section 3 are written separately. Ordinary dilation continuity, as in Kretschmann, Schlingemann and Werner [3, Definition 1 and Theorem 1], does not itself impose a comparison acting only on the active bath with an already fixed inert purifier and positive initial spectrum. This is a difference in the stated contract, not proof of originality of Theorem 4.1.

In particular, purifying a mixed initializer gives a dilation into ABF. The environmental comparison in ordinary Stinespring continuity can act on the joint BF; it need not factor as an operation on B tensored with I_F . Theorem 4.1 instead preserves the old positive spectrum at the support-repair stage and completes columns inside the target Kraus span, with a gain bound uniform in the bath size. The final flagged exactification changes the spectrum. These are the precise extra requirements in the comparison.

For factorization background, Haagerup and Musat [22, Definition 1.3 and Theorem 2.2]

characterize matrix Markov maps through unitary dilations with a tracial von Neumann algebra. Their finite von Neumann algebra need not be finite dimensional; our device bath must be. This is background to the closure/attainment distinction above, not a historical claim that they originated all factorization terminology.

The privacy-funnel comparison in Definition 1.3 identifies the exact pointwise cost, with its smoothing equality supplied by Section 4. The classical privacy funnel originates with Makhdoumi et al. [30]; Datta, Hirche and Winter supply the quantum definition used here. Other channel entropies answer different questions. In particular, Gour and Wilde [19, Proposition 6, Theorem 10 and Proposition 24] give a channel entropy and a universal parallel channel-merging theorem. Their protocol allows operations on both output and environment shares, free one-way classical communication, and measures net entanglement gain. Its channel entropy is $-\log q$ for a unitary and zero for a pure replacer; our κ is respectively zero and $\log q$. Their universal merging result is a close antecedent, but its resource accounting and access differ from ours.

Slepian–Wolf attribution is explicit in Section 5. Its encoder and the universal merging and reverse-Shannon constructions, including Bennett et al. [12, Theorem 3], provide the relevant splitting machinery. Our immediate-output construction runs exact collisions before encoding spent cells. It must additionally control adaptive spectra and allocate actual flat stock, seed, work and all residues in one closed bath; communication or net entanglement rates alone do not count that allocation.

Baghali Khanian and Leung [27, section II, Definition 3 and Theorems 6 and 8] strengthen the reverse-Shannon comparison by allowing a general mixed source/reference and preserved encoder-side information, unifying feedback and non-feedback simulation. Their input is a specified source tensor power, and the rates charge a transmitted register and entanglement. Their assisted optimum and unassisted bounds do not allocate all encoder/decoder environments inside one closed sequential bath. Transferring that machinery to Definition 1.1 still requires the adaptive compression, timing and complete storage accounting of Appendix A.

Devetak and Yard [25, main region and Eqs. (1)–(2)] place fully quantum Slepian–Wolf in state redistribution: the resources are quantum communication and net entanglement for many identical copies of a specified state with accessible sender and receiver shares. Those resource rates do not by themselves price the complete retained bath of Definition 1.1.

Faist and Renner [24, Main Result and Eq. (2)] characterize a specified process on a specified input, preserving its reference correlations, by coherent relative entropy with an information battery and Gibbs-preserving free operations. This is a work resource comparison; it does not impose our whole-bath dimension charge or one device serving all adaptive testers.

Faist, Berta and Brandão [20, Theorem 5.1 and Eq. (5.4)] give a universal parallel thermodynamic implementation with work rate $\max_{\rho}[S(\rho) - S(\Phi(\rho))]$ for trivial Hamiltonians. Their battery and free-operation accounting includes a partial-trace implementation, so it does not charge the entire retained bath of Definition 1.1. The two models are nevertheless comparable in one direction: Proposition 7.4 shows that our entropy-deficit rate $r - s$ is at least that work rate, for every channel. The mixed-bath model of Terhal et al. already imposes the unitary column constraints; its finite single-use environment analysis does not state this repeated rate region. Lie and Jeong [21, Theorems 1, 5–6] study randomness-utilizing implementations with input-independent bath output. That restriction is absent here, and their no-secret recovery acts on a purifier that our device cannot access.

Quantum strategy memory in Bisio et al. [13, Definitions 3–4 and Theorem 3] permits free classical memory and local CPTP implementation; Definition 1.1 charges those physical resources.

Lie and Jeong [29, Proposition 2, Corollary 10 and Theorem 11] study implementations returning a catalyst for every input. Their entropy bounds specialize to $s \geq h/2$ and $r + s \geq h$ for a returned catalyst implementing the tensor-power channel. Definition 1.1 imposes no such

return; its region also permits $s = 0$ and $r = h$. Their catalytic block decomposition does not identify every zero-gain collision: a SWAP with a maximally mixed bath has $g = 0$ but returns the input state to the bath. No reverse closure implication is imported here.

Kotowski and Kotowski [34] implement a unital channel on a d -dimensional system with a fresh ancilla of dimension k and success probability of order $k/\log d$, shown optimal up to constants, and simulate highly noncommutative channels with one auxiliary qubit. Their protocol may fail with a flag and draws a fresh ancilla at every use; Definition 1.1 removes both permissions, since the device must succeed at every visit with one bath that is never replenished, so neither cost measure bounds the other and nothing is imported.

Dimension and entropy need not have the same minimizer in quantum models of classical stochastic processes, as shown by Loomis and Crutchfield [31] and Liu et al. [32]. Those costs concern a stationary information-bearing memory; here actual initial entropy is priced together with the entire closed device. In the quantum-input setting, Chang, Berk and Gu [33, Theorem 3 and Appendix B.5] bound stationary recurrent memory by temporal excess entropy. Their bound vanishes for a memoryless channel, and their recurrent step may be an arbitrary CPTP operation. Definition 1.1 instead counts the entire closed apparatus needed to supply repeated calls. These comparisons explain the operational content of the dimension–initial-entropy characterization. Known entropy, minimax and decoupling ingredients do not by themselves supply the complete closed implementation; combining them under this contract is what the paper adds. None of the works cited above states a rate region for the closed contract of Definition 1.1.

9 Conclusion and open questions

Under the closed contract a single-use dilation question becomes a rate region in two resources, bath dimension per use and initial entropy per use, with h fixing their sum and κ fixing their difference. Three questions are left open here. Computability: κ is an ordered limit of an infimum over extensions of unbounded dimension, and no algorithm, finite witness bound or continuity in Φ is asserted; Proposition 7.4 gives only a lower bound that is easy to evaluate, and Proposition 7.5 leaves a gap of a factor of about three between that bound and the exhibited upper bound. Efficiency: the all-horizon device of Theorem 5.1 is built from a witness that is not constructed, and no circuit-size or latency claim is made; the streaming realization for dephasing shows that some corners admit explicit efficient devices, and which channels do is open. The contract: immediate output return, one visit per input and a memoryless target are all load-bearing here. Devices that deliver outputs on a schedule rather than immediately are studied in a companion note on Schur channels [35]; testers with bounded persistent quantum memory rather than unrestricted references are a separate question that this paper does not address.

AI assistance

AI systems assisted with proof exploration, manuscript preparation and checking. Additional independent AI sessions audited the arguments and their relationship to prior work. These sessions were not human external peer review. The scope of the separate, partial Lean verification is stated in Appendix B.4.

A Complete causal balancing proof

This appendix proves Theorem 5.1. Fix the channel and one finite *exact* collision before any limit. Use σ, v, g, h, a, b, R from Theorem 5.1, with $0 \leq a \leq \sigma, g \geq 0$ and $b = a + g$. All logs are

base two and D denotes half trace distance throughout; in Lemma A.3 the integer dimension of the compressed register is written D_C .

A.1 Adaptive spectral concentration

Lemma A.1. *Let Λ have finite output and $H_\Lambda = \max_\rho S(\Lambda(\rho))$. There is a fixed spectral projector on n successively produced, untouched Λ outputs with rank at most $2^{n(H_\Lambda + \gamma)}$ and rejected weight at most $\exp(-c_\Lambda n \gamma^2)$, uniformly over adaptive controllers and all finite reference dimensions. Deterministic spectral costs have zero loss.*

Proof. Choose an entropy-maximizing output ζ . Its support contains every output support: mixing an output with positive weight outside $\text{supp } \zeta$ into ζ would increase entropy by a positive leading $-t \log t$ term. On the common support the directional entropy derivative toward any output ω is $-\text{Tr}(\omega - \zeta) \log \zeta \leq 0$. Thus

$$\text{Tr } \Lambda(\rho)(-\log \zeta) \leq S(\zeta) = H_\Lambda.$$

Let $z_j > 0$ be its eigenvalues and $L_\Lambda = \max_j (-\log z_j)$. Hypothetically measure each spent output in this basis. Conditioned on earlier outcomes, the next input is a density operator and its channel resource is fresh; therefore $X_i = -\log z_{j_i}$ has conditional mean at most H_Λ and lies in $[0, L_\Lambda]$. More explicitly, for a positive-probability hypothetical history $j_{<i}$, let $\xi_{S_i K_i | j_{<i}}$ be the normalized conditional state just before visit i , with K_i containing the controller and all its references. Freshness makes the current dilation resource product with this joint state. Writing $\rho_i(j_{<i}) = \text{Tr}_{K_i} \xi_{S_i K_i | j_{<i}}$, one has

$$\Pr(j_i = j \mid j_{<i}) = \langle j | \Lambda(\rho_i(j_{<i})) | j \rangle, \quad \mathbb{E}[X_i \mid j_{<i}] = \text{Tr } \Lambda(\rho_i(j_{<i}))(-\log \zeta) \leq H_\Lambda.$$

The conditional input may depend on the full history and be mixed because of coherent entanglement; neither fact changes the channel-output formula. Zero-probability histories may be omitted. These measurements are proof devices, not information supplied to the controller. The conditional exponential-moment bound follows directly by iterating $\mathbb{E}[\exp(t(X_i - \mathbb{E}[X_i \mid \text{past}])) \mid \text{past}] \leq \exp(t^2 L_\Lambda^2 / 8)$, then minimizing $\exp(-tn\gamma + nt^2 L_\Lambda^2 / 8)$ over $t \geq 0$. It gives

$$\Pr\left\{\sum_i X_i > n(H_\Lambda + \gamma)\right\} \leq \exp(-2n\gamma^2 / L_\Lambda^2).$$

Indeed $\mathbb{E}[\exp(t(X_i - H_\Lambda)) \mid \text{past}] \leq \exp(t^2 L_\Lambda^2 / 8)$ for $t \geq 0$, which gives the iteration even when the conditional means vary. Thus $c_\Lambda = 2/L_\Lambda^2$ is independent of the tester; it need not be uniform over different channels or witnesses with small positive eigenvalues.

This is conditional iteration of the bounded-variable exponential-moment estimate underlying Hoeffding's inequality [17]. For completeness, the log moment-generating function of a variable in $[0, L]$ has second derivative equal to its variance in an exponentially tilted law, at most $L^2/4$. Integrating twice from zero gives the centered bound $t^2 L^2 / 8$, also for each conditional law. The application here selects one spectral cost from the entropy-maximizing output, uniformly over controllers.

For $L_\Lambda = 0$ the event is empty. The hypothetical measurements commute with subsequent service and controller operations up to the boundary just before compression, because these operations do not touch spent outputs. Thus the same probability is the rejected weight of the unmeasured state at that boundary. No commutation with the later compressor or encoder is asserted or needed. Every accepted eigenstring has product ζ weight at least $2^{-n(H_\Lambda + \gamma)}$, so there are at most $2^{n(H_\Lambda + \gamma)}$ such strings. This proves the claim without i.i.d. inputs or a union bound over users. $\square$

For $\Lambda = \Gamma_C$, fresh τ cells therefore give a user-independent P_B on the spent B^n cells with rank at most $2^{n(v+\gamma)}$ and $\text{loss}_B \leq \exp(-c_B n \gamma^2)$. All physical basis changes and compressors depend only on this fixed witness, n and γ . The same lemma applies to untouched minimal environments of Φ .

For comparison, Metger et al. [23, Theorem 4.1 and Appendix A] bound sequential smooth entropies. Their min-entropy theorem requires $\text{Tr}_{A_i R_i} M_i = \mathcal{R}'_i \text{Tr}_{R_{i-1}}$; Appendix A Eq. (A.2) there gives a max-entropy form without that restriction. Such statewise bounds alone do not specify one compression projector shared by every tester. Here the fixed P_B preserves all complete adaptive tests in Definition 1.2 through the trace-distance argument; Lemma A.2 additionally constructs simultaneous support and spectral-cap cuts. We use the direct proof above, not an entropy-accumulation citation, to infer projector uniformity or seed freshness.

A.2 The complete purified exterior

Lemma A.2. *In an exact comparison block there are commuting proof projectors on B^n, H, F^n whose success p is at least $1 - \text{loss}_B - \text{loss}_E - \text{loss}_F$, and whose normalized pure projection obeys*

$$\text{rank } \hat{\rho}_H \leq K = 2^{n(h+\gamma)}, \quad \|\hat{\rho}_{B^n H}\|_\infty \leq 2^{-n(\sigma-\gamma)}/p.$$

Here H is a purification of the complete future-accessed exterior, excluding the independent seed and current source. No rank bound on its raw mixed marginal is asserted.

Proof. At a comparison boundary let E_{raw} contain the entire user and references, all unused stock, and every other spectator that will be accessed again. Exclude the seed Z , the n current source cells, and permanently parked registers. The source $\tau^{\otimes n}$ is independent of E_{raw} and Z jointly. Purify E_{raw} to $H = E_{\text{raw}} E'$ for analysis, and purify each source with F_i . Dilate the tester's operations, retaining all its records in H . The final $B^n F^n H$ state is pure, and the F^n marginal stays $\tau_F^{\otimes n}$. The device acts on neither F^n nor E' . These are comparison purifications, not extra physical workspace or modifications of an actual inert purifier.

Run the same purified controller with fresh pure minimal dilations of Φ instead. The resulting virtual state $\theta_{E^n H}$ is pure and its H marginal equals the actual comparison H marginal: both provide exactly Φ at every visit, including to reference-entangled inputs. Lemma A.1 gives a virtual P_E of rank at most K and $\text{loss}_E \leq \exp(-c_E n \gamma^2)$. Define

$$X_H = \text{Tr}_{E^n} [(P_E \otimes \mathbb{I}) |\theta\rangle \langle \theta| (P_E \otimes \mathbb{I})].$$

Then $0 \leq X_H \leq \rho_H$, since partial trace kills the cross terms between P_E and $\mathbb{I} - P_E$, and $\text{rank } X_H \leq \text{rank } P_E \leq K$. If $P_H = \text{supp } X_H$, then $\text{Tr } P_H \rho_H \geq \text{Tr } X_H \geq 1 - \text{loss}_E$. The projector P_H may depend on the user and is only a proof projector. An initially mixed spectator may have arbitrarily large raw rank even when $h = 0$; including E' before this argument is essential.

The positive eigenvalues of τ have finite logarithmic costs. The ordinary lower-tail bound for independent source eigenstrings supplies P_F accepting $-\log p_x \geq n(\sigma - \gamma)$, with $\text{loss}_F \leq \exp(-c_F n \gamma^2)$, and

$$P_F \tau_F^{\otimes n} P_F \leq 2^{-n(\sigma-\gamma)} P_F.$$

Zero eigenvalues have zero probability and are omitted. Deterministic-cost tails have zero loss. The projectors P_B, P_H, P_F act on distinct registers and commute; the union bound gives the stated p . Crucially the virtual P_E is not applied to the mixed-collision purification alongside a physical bath projector.

For every vector on F , the quadratic form of the F marginal after projecting BH is at most its original value, because $P_B P_H$ is a positive contraction. Thus that unnormalized F marginal is at most $\tau_F^{\otimes n}$. After P_F and normalization it is capped by $2^{-n(\sigma-\gamma)}/p$. Purity transfers this nonzero spectrum to $B^n H$. The normalized projected state has half distance $\sqrt{1-p}$ from the original pure state. No independence of spectral events is needed, and the factor $1/p$ cannot be suppressed. $\square$

A.3 Second moment with a retained physical seed

Lemma A.3. *Let $C = MG$, $\dim C = D_C$, $\dim M = m$, and ρ_{CH} be normalized with $\text{rank } \rho_H \leq K$ and $\|\rho_{CH}\|_\infty \leq \lambda$. A uniform exact unitary two-design V_z on C gives*

$$\mathbb{E}_z \|\text{Tr}_G(V_z \rho_{CH} V_z^*) - \pi_M \otimes \rho_H\|_1 \leq \sqrt{m^2 K \lambda / D_C}.$$

If an independent actual classical seed Z is retained, the half trace distance of ZMH from $\pi_Z \otimes \pi_M \otimes \rho_H$ is at most half this expression.

Proof. For $\Delta_V = \text{Tr}_G(V \rho_{CH} V^*) - \pi_M \otimes \rho_H$, the two-copy Haar twirl of $\text{Swap}_M \otimes \mathbb{I}_{GG}$ equals $\alpha \mathbb{I} + \beta \text{Swap}_C$. The untwirled operator has trace D_C^2/m and trace against Swap_C equal to $D_C m$; hence, for $D_C > 1$,

$$\beta = \frac{D_C(m^2 - 1)}{m(D_C^2 - 1)} \leq \frac{m}{D_C}, \quad \alpha = \frac{1}{m} - \frac{\beta}{D_C}.$$

The swap trick and the cross term against $\pi_M \otimes \rho_H$ give exactly

$$\mathbb{E} \text{Tr } \Delta_V^2 = \beta \left[\text{Tr } \rho_{CH}^2 - \text{Tr } \rho_H^2 / D_C \right].$$

Δ_V is supported on $M \otimes \text{supp } \rho_H$, of dimension at most mK . Cauchy–Schwarz for its singular values, Jensen, and $\text{Tr } \rho_{CH}^2 \leq \lambda$ give the claimed trace-norm bound. If $m = 1$, then $\Delta_V = 0$ exactly, including $D_C = 1$. An exact two-design has these same second moments. This is the elementary specialization underlying Dupuis et al. [5, Theorem 3.3 and Lemmas 3.4–3.5]; the factor 1/2 is explicit here.

For N qubits, the uniform Clifford group modulo global phases is such a finite exact two-design, by Dankert et al. [6, Theorem 1, pp. 2–3]. Its size J_N has $\log J_N \leq 2N(2N+1)$: each of $2N$ Pauli generators has at most 2^{2N+1} signed images, and these images determine conjugation. For $N = 0$ use $J_0 = 1$. Initialize Z in π_{J_N} offline and apply $\sum_z |z\rangle\langle z| \otimes V_z$. If Z is independent of CH *jointly* beforehand, the cq block trace-norm identity gives

$$D(\rho_{ZMH}, \pi_Z \otimes \pi_M \otimes \rho_H) = \frac{1}{2} \mathbb{E}_z \|\Delta_{V_z}\|_1.$$

Keeping the random unitary choice in the output is also explicit in Dupuis et al. [5, discussion after Corollary 3.2]. The additional accounting here charges this physical label and reuses it across blocks through the joint comparison in Proposition A.5.

No one seed value is selected for all users. This is a guarantee for each user on a physically retained mixed seed, with constants uniform over users. The seed’s inert purifier is not included in this reduced independence claim; it is never accessed. Marginal independence of Z from C and from H separately would not suffice. $\square$

A.4 Exact integer wires and reversible preparation

Lemma A.4. *Fix $n \geq 1$ and $\gamma > 0$. Define integer widths*

$$\begin{aligned} N &= \lceil n(v + \gamma) \rceil, & m_i &= \lceil n(\sigma + 2\gamma) \rceil, & m_o &= \max(0, \lfloor n(\sigma - a - 6\gamma) \rfloor), & r_p &= \lceil 3n\gamma \rceil + 4, \\ c &= m_i - m_o, & d &= N + r_p - m_i, & z &= c + d = N - m_o + r_p, \\ e &= n \lceil \log R \rceil, & w &= \max(0, e - N). \end{aligned}$$

The full block can be implemented on permanent M, W of m_o, w qubits and one disjoint z -qubit tranche. The tranche starts with c mixed and d pure qubits. Both preparation and encoding residues end in that tranche, which is never accessed again. All dimensions in this statement are physical.

Proof. The typical-spectrum threshold is the usual source-coding ingredient; see Abeyesinghe et al. [7, section VII and Appendix A]. Here the preparation is implemented as an injection of flat labels with all its fibres retained on the counted wires below. Zero-pad the R -dimensional cell to $\lceil \log R \rceil$ qubits and extend U unitarily outside its valid sector. From $0 \leq a \leq \sigma$, $g = v - \sigma \geq 0$ and $v \leq \log R$, we have $c, d \geq 0$, $m_o \leq m_i$, $m_o \leq N$, $m_i \leq e + r_p$ and $z \geq r_p$. In particular $d \geq ng + 2n\gamma + 3$ and $m_i \leq e + 2n\gamma + 1 \leq e + r_p$. Clipping m_o changes its ideal leading term $n(\sigma - a)$ by at most $6n\gamma + 1$. Consequently, uniformly over clipping,

$$c = na + O(n\gamma + 1), \quad d = ng + O(n\gamma + 1), \quad z = nb + O(n\gamma + 1).$$

At opening, M and the fresh c mixed tranche qubits give exactly m_i mixed qubits, while all other local wires are pure. Their exact volume identity is

$$m_i + d + w = N + r_p + w = e + r_p + \max(0, N - e) = m_o + z + w. \quad (\text{A.1})$$

In the eigenbasis of $\tau^{\otimes n}$, for strings with probabilities in $[2^{-n(\sigma+\gamma)}, 2^{-n(\sigma-\gamma)}]$, assign $\lfloor 2^{m_i} p_x \rfloor$ distinct flat input labels to cell label x and different preparation-residue labels. Each fibre uses at most $2^{3n\gamma+1} < 2^{r_p}$ labels. There are at most $2^{n(\sigma+\gamma)}$ typical strings, so their total rounding deficit is at most $2^{n(\sigma+\gamma)-m_i} \leq 2^{-n\gamma}$. The desired subdistribution is pointwise below the target distribution. Send leftover labels injectively to unused cell/residue pairs; $m_i \leq e + r_p$ guarantees capacity. Although these labels can raise some marginal probabilities above target, total variation is at most the missing subdistribution mass, giving

$$\delta_{\text{prep}} \leq \Pr\{\text{source atypical}\} + 2^{-n\gamma}.$$

The two-sided source tail is at most $2\exp(-c_\tau n\gamma^2)$, with deterministic costs treated exactly. The injection of an orthonormal input set extends to a unitary on the full local space in (A.1); any $\max(0, N - e)$ padding stays pure. The complete local positive spectrum remains exactly 2^{m_i} copies of 2^{-m_i} . Only its cell marginal is approximately nonflat.

Route the r_p preparation-residue positions into the current tranche and never touch them again. In the comparison run replace only the cell marginal by $\tau^{\otimes n}$; all future-accessed exterior registers remain independent. The preparation error is charged once for the entire block by contractivity, even with adaptive service. This is a comparison of reduced states, not a physical erasure or operation on the purifying systems.

Run each exact collision when its input arrives. At an interior visit, hand back its output after that collision. At the final visit of a block, perform the collision, then the bath-only closing gates described below, and then release that visit's output, all before the next input arrives. Bath-only unitaries preserve the joint reduced state of every user output and reference, including the current output. On e cell positions and $\max(0, N - e)$ padding, a fixed unitary compresses $\text{supp } P_B$ into N qubits C and w pure qubits W . The rank bound in Lemma A.1 guarantees this embedding; complete its remaining columns orthonormally. No projection is performed or success flag measured. Split C into M of m_o qubits and G of $N - m_o$ qubits, use Lemma A.3's controlled encoder, and route M, W to their permanent positions. Park G beside the preparation residue in the same tranche. The equality

$$z = r_p + (N - m_o) \quad (\text{A.2})$$

proves that neither residue overlaps the other or any future stock. The compression failure part remains coherently present on these same wires. $\square$

A.5 Joint recycling, every prefix and the repeated unitary

Proposition A.5. *The block above has a user-independent reduced-state error $\epsilon_n \leq A[\exp(-c_0 n\gamma^2) + 2^{-c_1 n\gamma}]$ for large $n\gamma^2$, returning M, W and the seed jointly independent of the entire future-accessed exterior in the comparison state. Constants depend only on the fixed collision.*

Proof. Apply Lemma A.2 to the exact comparison block. After compression its projected W is pure and its normalized CH state has rank bound K and cap $\lambda = 2^{-n(\sigma-\gamma)}/p$. Before encoding the seed is independent of CH jointly: preparation and service have not touched it, and the proof cuts can be chosen without referring to it. In particular the comparison input is $\pi_Z \otimes \rho_{\text{source,exterior}}$, with source and exterior also in product. Its purification and P_H may depend on the tester but not on Z ; P_B depends only on the fixed collision. Applying these proof cuts therefore preserves the product with π_Z . This is the joint hypothesis of Lemma A.3, including all unused stock and future-accessed spectators in the exterior. For $m_o > 0$,

$$2m_o + n(h + \gamma) - n(\sigma - \gamma) - N \leq -11n\gamma,$$

since $\sigma - 2a + h - v = 0$. Lemma A.3 therefore bounds the projected joint half distance by $2^{-11n\gamma/2}/(2\sqrt{p})$. For $m_o = 0$ it is zero. Add $\sqrt{1-p}$ for smoothing the initial pure comparison, and another for replacing its projected H marginal by its original H marginal. W is pure on the projected branch, so it is included jointly in the same estimate. Tracing the mathematical E' then gives the bound on the physical exterior. Including preparation once yields

$$\epsilon_n \leq \delta_{\text{prep}} + 2\sqrt{\text{loss}_B + \text{loss}_E + \text{loss}_F} + \frac{2^{-11n\gamma/2}}{2\sqrt{p}} \quad (\text{A.3})$$

when $m_o > 0$; omit the last term otherwise. With $p \geq 1/2$ the asserted bound follows, absorbing square roots into fixed positive constants.

For L full blocks the boundary comparison is

$$\rho_{\text{user,ideal}} \otimes \pi_Z \otimes \pi_M \otimes |0\rangle\langle 0|_W \otimes (\text{all unused mixed/pure tranches}) \otimes (\text{pure tail cells}). \quad (\text{A.4})$$

Old tranches are omitted only from this trace-distance metric; their full dimensions remain allocated. Initially the actual reduced state is (A.4), so $e_0 = 0$. Fix any adaptive tester, apply the *same* next-block CPTP map to its actual state and to (A.4), and contract their distance. The uniform local bound then gives $e_{j+1} \leq e_j + \epsilon_n$. Thus actual learning of the seed and correlations with recycled M are charged to e_j . Freshness is used only in the comparison run, where the next source and seed are jointly independent of the complete exterior. There is no assumption about independence of errors.

At a prefix inside a block before its closing encoder, the user error is at most $e_j + \delta_{\text{prep}}$. The closing bath-only gates can be performed after the last collision and before handing back that visit's output, preserving immediate service. Fewer than n tail visits use preallocated pure minimal-Stinespring cells, exactly implementing Φ ; these are part of the initial bath. Thus the entire horizon and every prefix obey the conservative bound $\delta_T \leq L\epsilon_n + \delta_{\text{prep}}$. When the final segment consists only of pure tail cells, no extra preparation term is needed. Stopping is handled by extending a stopped tester with dummy inputs while keeping its records and stop flag, then tracing the continuation. This is unconditional; no rare-event postselection estimate follows.

All opening, collision, closing and routing steps for visit t form a fixed unitary V_t on the system and allocated non-clock bath. A pure counted $(T+1)$ -dimensional clock implements the *same* unitary each visit:

$$W_T = \sum_{t=0}^T |t+1 \bmod (T+1)\rangle\langle t|_{\text{clock}} \otimes V_t. \quad (\text{A.5})$$

Choose the unused V_T arbitrarily unitary. Orthogonality of clock sectors and unitarity of each V_t prove W_T is unitary. Starting at 0, the first T visits do not wrap. All control resides in this fixed hardware. There is no bound on gate count or latency, but the next input never arrives before the previous output is handed back. $\square$

A.6 Actual spectrum and the every-horizon limit

Proposition A.6. *The construction has the exact initialization and separate limits asserted in Theorem 5.1.*

Proof. Set $L = \lfloor T/n \rfloor$ and $t_{\text{tail}} = (T - Ln)\lceil \log k \rceil$, where $k = \text{rank } J_{\Phi}$ is the minimal environment dimension. Zero-padding each pure tail cell and unitary completion are counted. The exact numbers of allocated data qubits and initially mixed data qubits are

$$A_T = m_o + w + Lz + t_{\text{tail}}, \quad K_T = Lc + m_o. \quad (\text{A.6})$$

Since $d \geq 0$, $K_T \leq A_T$. With the clock and the one reused seed,

$$\begin{aligned} R_T &= (T + 1)J_N 2^{A_T}, & \omega_T &= |0\rangle\langle 0|_{\text{clock}} \otimes \pi_{J_N} \otimes \pi_{2^{K_T}} \otimes |0\rangle\langle 0|_{\text{rest}}, \\ P_T &= J_N 2^{K_T}, & S(\omega_T) &= \log J_N + K_T. \end{aligned} \quad (\text{A.7})$$

Its positive eigenvalues are exactly $1/P_T$ repeated P_T times, with $R_T - P_T$ zeros. Its minimal inert purifier has dimension P_T and is never accessed. The witness entropy σ is an intermediate preparation parameter; it is not substituted for the actual entropy in (A.7).

For every sufficiently large integer T take $n = \lfloor T^{1/3} \rfloor$ and $\gamma = n^{-1/4}$. Fixed-witness constants have already been chosen. Then $n\gamma^2 = \sqrt{n}$, so the exponential in (A.3) beats the polynomial number L of blocks; the other exponent is of order $n^{3/4}$. Thus

$$L\epsilon_n + \delta_{\text{prep}} \rightarrow 0, \quad A_T/T \rightarrow b, \quad K_T/T \rightarrow a, \quad \log J_N/T = O_C(n^2/T) \rightarrow 0,$$

$$(m_o + w + t_{\text{tail}})/T = O_C(n/T) \rightarrow 0, \quad \log(T + 1)/T \rightarrow 0.$$

Slack costs $O(T\gamma)$, rounding costs $O(T/n)$, and the seed $O_C(n^2)$ bits; all are sublinear. Finitely many smaller horizons use preallocated exact pure cells and a counted clock. Consequently for every fixed witness and every tolerance there is a finite threshold after which *every* horizon has small error and small rate discrepancies, including the separate seed and mixed-data limits. No effective or witness-uniform threshold is asserted.

The same integers handle $m_o = 0$, $g = 0$, $a = 0$, $R = 1$ and $N > e$. For example pure τ forces $\sigma = a = 0$; its mixed slack consumption is $o(T)$. An identity collision with an arbitrary mixed spectator has $g = h = a = b = 0$ and its startup stock is $O_C(n)$. These checks use no negative register widths. $\square$

The argument uses encoder-only fully quantum Slepian–Wolf, with its would-be message parked and counted; see Section 5 for attribution. All original proof steps needed beyond that ingredient have been supplied above. This concludes the proof of Theorem 5.1.

B Conventions, resources and references

This appendix records conventions, registers, reproducible finite checks and the bibliography. It adds no claims.

B.1 Normalization and finite objects

Symbol	Convention
S, A, Q	Input, immediate output and purification reference; dimension q each.
ρ_Q	ρ^T for the canonical purification $(\mathbb{I} \otimes \sqrt{q\rho})\Omega_q$; spectra and minimum eigenvalues agree with ρ .
J_Φ	Trace-one Choi state, $\text{Tr}_A J_\Phi = \mathbb{I}_Q/q$; $k = \text{rank } J_\Phi \leq q^2$.
$D, D_\diamond$	Half trace norm and half diamond norm; range $[0, 1]$ for states and channels.
S, H_2	Base-two von Neumann and binary entropy; $0 \log 0 = 0$.
π_d	$\mathbb{I}_d/d$, with d an actual integer Hilbert dimension.
$k_{\rho, \varepsilon}$	All normalized finite-QAZ extensions with exact QZ product and QA error at most ε ; auxiliary dimension unrestricted.
κ	Full-rank input supremum after the positive-error limit at fixed ρ ; Corollary 4.3 derives the exact privacy-funnel form using Theorems 3.1 and 4.1, without attained witnesses.
$K_0(\rho), P_q$	Exact fixed-input extension infimum and quantum privacy-funnel supremum with $X = E, Y = Q, R = A, W = Z$; unrestricted finite auxiliary dimension. The prior subscript q means quantum.
Γ_C	Active bath output only; no inert purifier included.
r, s	Actual limits $\log R_T/T$ and $S(\omega_T)/T$ for vanishing-error all-horizon families.
Entropy deficit	$\log R_T - S(\omega_T)$; neither entropy nor log rank.
Appendix A constants	May depend on a single fixed finite exact collision; uniform in tester, finite reference dimension and horizon.
Reused letters	F is the inert purifier of the actual initializer (Definition 1.1) and, as F^n in Appendix A, the purifier of the comparison stock; the Slepian–Wolf receiver is written N . C names both a finite collision (U, τ) and the Slepian–Wolf sender register C . k is the Choi rank; $k_{\rho, \varepsilon}$ is the smoothed extension cost. ν in Proposition 7.5 is a damping parameter; γ in Appendix A is the concentration slack.

The case $q = 1$ is scalar and is directly realized without a bath. In Theorem 4.1 one uses the smallest *positive* target Choi eigenvalue, never a kernel inverse. Square input and output dimensions are needed for the support-isometry unitary completion in Theorems 3.1 and 4.1. All finite internal direct sums and zero padding are included in the active dimension.

B.2 Physical and proof registers of Appendix A

Register	Size / initialization	Action and charge
Full B_T	$R_T = (T + 1)J_N 2^{A_T}$	Every active device register below is included.
Clock	$T + 1$; pure	Cyclic shift controls the fixed per-visit unitaries; $\log(T + 1)$ charged.
Z	J_N ; π_{J_N}	One retained seed reused for every block; $\log J_N$ charged in dimension and entropy.
M	m_o qubits; maximally mixed	Permanent recycled capital, returned jointly in the reduced comparison.
W	w qubits; pure	Permanent workspace; returned pure jointly up to the block error.
Tranche j	$z = c + d$ qubits; c mixed and d pure	Distinct for every full block. Preparation residue r_p and encoding residue $N - m_o$ fill it permanently.
Working cells	$e = n \lceil \log R \rceil$ positions during a block	Obtained by routing within the exact local wires, not a separate free bank.
Compressor padding	$\max(0, N - e)$ pure positions	Included in identity (A.1); handles $N > e$.
Tail	$(T - Ln) \lceil \log k \rceil$ pure qubits	Preallocated exact pure-Stinespring cells; retained after use.
Actual F	Minimal dimension $J_N 2^{K_T}$	Purifies the complete initializer and is forever inert; not an active resource.
Proof F^n	Purifies comparison $\tau^{\otimes n}$	Not the actual repeated stock initializer and never an operated register.
$H = E_{\text{raw}} E'$	Purification of the complete comparison exterior	Includes all future-accessed spectators before its rank bound; E' is mathematical only.
Virtual E^n	Fresh minimal environments in a comparison	Produces support projector P_H ; virtual P_E is never a physical compression operation.

Only the reduced boundary distance omits parked tranches and inaccessible purifiers. Their omission from a metric does not delete any actual bath dimensions. Exact initialization is P_T equal eigenvalues $1/P_T$, where $P_T = J_N 2^{K_T} \leq R_T$; its remaining $R_T - P_T$ eigenvalues are zero.

B.3 Scope of the statements

The physical contract for every statement in this paper is Definitions 1.1–1.4. Nothing above uses a theorem about finite-tracial closure, about the strength of a physical record, about feedback-assisted protocols, about computability, or about processes with more than one visit per input. Propositions 7.1–7.5 and 8.1 are proved here under those same definitions. No external classification of generalized amplitude damping, arbitrary mixed replacers or unequal pure-output sectors is used.

Four shortcuts are deliberately avoided and are used nowhere above: raw mixed-history rank in place of purified-exterior support, merely marginal seed freshness, uniform initial-entropy continuity under a small flag, and attained witnesses.

B.4 Reproducible diagnostics and their limits

The scripts named below accompany this paper in `ancillary/`; run the Python commands from that directory. In the repository they are in `verification/python/`. For Lean, use the wrapper command in `ancillary/COVERAGE.md` with an already installed compatible package cache; the table names its target files, not standalone commands with configured imports. They print their results; all are bounded finite controls. They do not prove unbounded auxiliary optimizations,

arbitrary adaptive asymptotics, or any theorem of this paper. Seeds and dimensions are explicit in the scripts.

Command in the ancillary bundle	Finite coverage
<code>python c2_lifting_mixing.py</code>	24 lifts, 20 flagged probes; $q = 2, 3$, $R = 2, 3, 4$; seed 20260908; tolerance 10^{-8} .
<code>python c3_exactification.py</code>	8 cases, 40 probes; $q = 2, 3$, original $R = 5, 7$; largest unitary 93; seed 20260909; tolerance 10^{-8} .
<code>python c4_adaptive_spectral.py</code>	8 nonproduct distributions, 112 conditional nodes; $q = R = 2$, $n = 3$; vector dimension ≤ 1024 ; seed 20260909; tolerance 10^{-8} .
<code>python c5_region_spectrum.py</code>	12 spectrum/feedback cases, 576 rational cases; seed 20260909; tolerance 10^{-9} ; density dimension ≤ 768 .
<code>python support_repair.py</code> <code>python block_invariant.py</code>	12 small-matrix cases; seed 20260908; tolerance 10^{-8} . 300 integer allocations, 1024-label preparation, joint-freshness and rare-stop negative controls, 12 states over 24 Cliffords.
<code>python encoder_only_merging.py</code>	36 partial-trace decoder-omission identity checks and 125 rational balances; seed 20260908; no decoupling or compiler validation.
<code>python rz_contract_negative_control.py</code> <code>python gad_interior_example.py</code>	Exact rational $q = R = 2$, $T = 1, \dots, 12$ marginal/joint separation; no random seed or numerical tolerance. Proposition 7.5: trace preservation, exchange-block unitarity, 2007 inputs reproducing the channel, both analytic endpoints, the exact rational lower witness, the attained upper witness, the mixed-input entropy-exchange value and diagonal grid searches for both; seed 20260910; tolerance 10^{-8} .
<code>BlockAccounting.lean</code> (via documented wrapper)	Six elementary natural-number accounting lemmas only.
<code>ActiveBath.lean</code> (via documented wrapper and pinned mathlib)	Finite complex density matrices and exact bath-only closing invariance for the complete user marginal.

The Lean files report their axioms and use only standard foundations (`propext`, `Quot.sound` and `Classical.choice`), with no `sorry` or added scientific axiom. `ActiveBath` defines positive-semidefinite trace-one complex matrices, proves that partial trace and unitary closing preserve legal states, and proves exact invariance of the complete user marginal under a bath-only unitary, including arbitrary user-bath correlations and finite references. This supports only the closing identity in Appendix A.4, not construction of the compiler. Coverage is *partial*: entropy, trace distance and the asymptotic theorem are not formalized. No numerical telescope suite for Theorem 2.2 is claimed. The ancillary coverage matrix and pinned dependency record give exact reproduction details.

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